

Do banks price firms' biodiversity risk? Evidence from the Kunming declaration

Jun Li^a, Yihang Jin^b, Peng Zhou^{c,*}

^a School of Economics, Hefei University of Technology; Institute for Carbon Neutrality and Finance, Hefei University of Technology

^b School of Economics, Hefei University of Technology

^c School of Economics, Hefei University of Technology; Institute for Carbon Neutrality and Finance, Hefei University of Technology, 485 Danxia Road, Hefei 230601, China

ARTICLE INFO

Keywords:

Biodiversity risk
Syndicated loans
Debt-paying ability
Green bank
Double materiality

ABSTRACT

The pricing of firms' biodiversity risk exposure, as an emerging dimension of environmental risk, remains a contentious topic. This study employs the international syndicated long loan data from 2013 to 2023 and adopts the exogenous shock of the Kunming Declaration as an identification strategy to examine the effects of biodiversity risk on firms' loan spreads. The findings reveal that: (1) Following the Kunming Declaration, firms with higher biodiversity risk are subject to significantly higher loan spreads, underscoring the presence of a biodiversity risk premium; (2) Firms with weaker debt-paying ability experience even greater increases in loan spreads when exposed to biodiversity risk. Additionally, banks with commitments to ecological sustainability, namely green banks, charge higher loan spreads to firms with higher biodiversity risk than other banks; (3) Banks weigh biodiversity transition risk more heavily than physical risk in their loan pricing processes; (4) From the perspective of the interaction between legal institutions and financial mechanisms, this paper finds that the biodiversity risk premium in syndicated loans is primarily driven by the common law system; (5) In addition to charging higher loan spreads, banks mitigate their biodiversity risk exposure by cutting loan amounts, shortening maturities, and raising fees. This study contributes new evidence to the ongoing controversy on biodiversity risk pricing and offers policy implications for policymakers, financial institutions, and firms in managing biodiversity risk.

1. Introduction

The most compelling and recent evidence indicates that global biodiversity is declining at an unprecedented rate (Johnson et al., 2024), driven by a multitude of adverse factors, including climate change and environmental pollution (Yang et al., 2024), habitat destruction and loss (Cabernard, Pfister, & Hellweg, 2024), as well as overexploitation and resource depletion (Lampert, 2019). Between 1970 and 2020, global freshwater, terrestrial, and marine species populations declined by 85 %, 69 %, and 56 %, respectively (WWF, 2024). A novel environmental risk—biodiversity risk¹—is emerging, attracting significant attention from policymakers, academics, and firm managers (Cherief, Sekine, & Stagnol, 2025; Fu, Steckbauer, Mann, & Duarte, 2024). Evidence has already shown that exposure to biodiversity risk can lead to significant

challenges for firms, such as rising operational expenses and escalating environmental compliance costs, ultimately threatening firms' survival and growth (Giglio, Stroebe, & Zeng, 2023). Furthermore, biodiversity risk increasingly exposes financial institutions' assets to biodiversity-related vulnerabilities, posing potential threats to the stability of the financial system (Hadjilazaro, Salin, Svartzman, et al., 2024). Existing literature primarily focuses on the impact of biodiversity risk exposure on firms' operational performance (Bach, Hoang, & Le, 2025; Li, Liu, Canil, & Cheong, 2025) and strategic decision-making (Tian & Chen, 2025; Trinh, 2023; Wang, 2025), while limited research examines whether biodiversity risk is recognized and priced by financial institutions. Drawing on syndicated loan data and exploiting the exogenous shock induced by the Kunming Declaration, this paper seeks to further identify the causal relationship between biodiversity risk and

* Corresponding author.

E-mail addresses: economics-lijun@hfut.edu.cn (J. Li), 2023171771@mail.hfut.edu.cn (Y. Jin), zhoupeng0414@163.com (P. Zhou).

¹ This primarily refers to the potential likelihood of biodiversity—encompassing species, genetic, or ecosystem diversity—being disrupted or lost due to human activities or natural factors.

bank loan spreads.

As a crucial source of external financing for firms, banks both provide credit financing services and participate in firm capital operations, such as equity investments, mergers, and acquisitions. According to microeconomic banking theory (Diamond, 1984), banks often function as delegated monitors, utilizing various tools to legally oversee borrowers' creditworthiness and behavior. This monitoring function serves to effectively mitigate issues of information asymmetry and moral hazard. At the same time, banks' client portfolios encompass diverse industries that are particularly vulnerable to biodiversity risk, including agriculture, manufacturing, and energy (Berger, Hasan, & Zhou, 2010). This extensive industry exposure facilitates banks' access to industry-specific information and data on biodiversity risk.

Moreover, with growing attention from governments, the public, and investors toward environmental issues, banks are under increasing pressure to shoulder greater social responsibilities. Beyond fulfilling their economic functions, banks are also expected to enhance the identification and management of environmental risks (Huang, Kerstein, Wang and Wu, 2022). As central actors in the financial system, banks play a pivotal role in directing capital flows and optimizing resource allocation. Consequently, their ability to identify, integrate, and disclose biodiversity risk exposures within their loan portfolios will shape the effectiveness and sustainability of biodiversity governance initiatives (Mundaca & Heintze, 2024). Whether banks effectively price biodiversity risk constitutes the first and most critical step in addressing such risks. However, existing literature has paid limited attention to the pricing of biodiversity risk in syndicated loans. Furthermore, current findings on pricing outcomes remain contested, and the underlying pricing mechanisms are still insufficiently explored (Becker, Di Girolamo, & Rho, 2025; Erten & Ongena, 2024). This underscores the urgent need for empirical validation, particularly through quantitative studies, to assess how biodiversity risk is integrated into banks' decision-making processes.

The publication of the Kunming Declaration provides a unique and valuable opportunity to explore the issues outlined above. Adopted during the 15th Conference of the Parties (COP15) to the Convention on Biological Diversity in Kunming, China, in 2021, the declaration establishes key global objectives and frameworks for biodiversity conservation. The Kunming Declaration is widely regarded as having significantly impacted global biodiversity risk, and is thus frequently considered an ideal quasi-natural experimental setting (Garel, Romec, Sautner, & Wagner, 2024; Kalhor & Kyaw, 2024; El et al., 2025). On the one hand, the declaration encourages countries to establish effective systems of protected areas to prevent the overexploitation of natural resources, such as oceans and land. This could directly constrain certain firms' access to natural resources, thereby impacting the regular operations of resource-dependent firms. On the other hand, the declaration calls on nations to enhance biodiversity-related legislation and strengthen environmental enforcement at the national level. As a result, firms with a substantial Corporate Biodiversity Footprint (CBF)² are likely to face heightened regulatory scrutiny and elevated compliance risks (Garel et al., 2024). Therefore, the issuance of the Kunming Declaration has posed more severe challenges for firms with high biodiversity risk exposure. The study uses the Kunming Declaration as an external shock to firms' biodiversity risk and employs a difference-in-differences (DiD) model to analyze whether biodiversity risk influences bank loan pricing.

This study identifies firms with significant biodiversity risk exposure

as the treatment group, while those with minimal exposure are assigned to the control group. The data on firms' biodiversity risk exposure is sourced from Coqueret, Giroux, and Zerbib (2025). Compared to methods that focus solely on measuring firms' impacts on the ecological environment (Garel et al., 2024), Coqueret et al. (2025) adopt a dual materiality framework that considers both firms' impacts on the ecological environment and the reciprocal effects of the environment on firms. This approach provides a more comprehensive and accurate measurement of biodiversity risk exposure.

Biodiversity risk is inherently long-term and cumulative, with its effects on firm performance typically emerging gradually over extended time horizons (Adamolekun, 2024). This characteristic may prompt banks to place greater emphasis on firms' exposure to biodiversity risk when pricing loans, particularly in the context of long-term lending. Accordingly, the long-term loan market serves as a critical setting for examining whether and how banks price biodiversity risk. This study centers on long-term bank lending and conducts an empirical analysis using international syndicated loan data from 2013 to 2023. The findings reveal that, first, following the Kunming Declaration, banks impose higher loan spreads on firms with greater exposure to biodiversity risk. Second, the weaker a firm's debt-paying ability, the more sensitive banks become to the firm's biodiversity risk exposure, demanding an additional risk premium as compensation. In addition, banks that claim to prioritize environmental sustainability and promote green development tend to impose higher loan spreads on firms with higher levels of biodiversity risk exposure.

Third, this research further distinguishes biodiversity risk into physical risk and transition risk. The former reflects firms' dependence on ecological resources for their daily operations, where the deterioration of ecosystems may expose them to risks such as disrupting operations, rising costs, and declining revenues (Giglio et al., 2023). The latter refers to risks associated with heightened public environmental awareness and strict regulatory frameworks. Firms with significant CBF that fail to adapt promptly may face reputational damage and increased compliance costs (Coqueret et al., 2025). The findings indicate that banks place greater emphasis on biodiversity transition risk, as evidenced by a higher sensitivity of loan spreads to such risks.

Fourth, this paper examines how cross-country differences in legal systems influence banks' pricing of biodiversity risk. The study finds that banks exhibit greater sensitivity to biodiversity risk in common law countries. The common law system emphasizes market discipline and contract enforcement (Ergungor, 2004). Within this institutional framework, banks rely more heavily on firms' internal environmental governance to assess their sustainability, making banks more inclined to incorporate biodiversity risk into loan pricing. By contrast, civil law countries typically feature stronger administrative regulation, which tends to weaken banks' ability or incentives to identify and price biodiversity risk. The findings underscore the indispensable role of legal institutions in the practice of green finance.

Finally, in addition to examining loan spreads, this study also examines the non-price terms of loan agreements. The results show that banks address the credit risks linked to borrowers' biodiversity risk exposure by lowering loan amounts, shortening maturities, and raising fees. These findings align with prior research by Fard, Javadi, and Kim (2020) and Javadi and Masum (2021), which suggest that when banks extend loans to firms affected by environmental risks, non-price contractual terms may be employed as alternative or supplementary mechanisms to reduce the banks' risk exposure.

This paper makes three potential marginal contributions:

Firstly, this study provides empirical evidence on the pricing of biodiversity risk, a novel environmental risk, from the perspective of bank lending. Current research on biodiversity risk largely focuses on its impact on firm operations and financial performance, such as operational performance (Bach, Hoang and Le, 2025), cash holdings (Ahmad & Karpuz, 2024), bankruptcy risk (Liang, Yang, Shen, & Dong, 2024), and stock price crashes (Bassen, Buchholz, Lopatta, & Rudolf, 2024).

² The Corporate Biodiversity Footprint (CBF), developed by Iceberg Data Lab (IDL), evaluates the extent of a company's effect on biodiversity. It assesses yearly biodiversity loss attributed to corporate activities by analyzing environmental factors such as land usage, greenhouse gas emissions, air pollution, and water contamination. This measure demonstrates the degree to which business operations disrupt ecosystems from their original condition.

However, few studies explored the effects of biodiversity risk on firms' borrowing costs. Leveraging the Kunming Declaration as an exogenous shock, this paper identifies the causal effect of biodiversity risk on loan spreads. Our main findings align with those of Coqueret et al. (2025) and Cherief et al. (2025), who analyze how biodiversity risk influences the cost of equity and bonds, providing a broader perspective on its negative impact on firms' capital costs.

Secondly, the findings of this study reveal that the weaker a firm's debt-paying ability, the more sensitive banks are to the firm's biodiversity risk exposure. Banks that have made green commitments fulfill their pledges to prioritize ecological sustainability and promote green development by demanding higher loan spreads from firms with high biodiversity risk exposure. The findings offer empirical support for policymakers in advancing the establishment of a green financial system. This can be achieved through measures such as establishing performance incentives for green banks and strengthening low-interest refinancing programs for green banking initiatives.

Finally, drawing on a dual materiality perspective, this study reveals the distinct effects of biodiversity physical and transition risks on loan pricing. The existing literature typically categorizes environmental risks into physical, transition, and regulatory risks (Shears, Meckling, & Finnegan, 2025), among which regulatory and transition risks are interrelated yet conceptually distinct dimensions.³ Currently, within the domain of environmental risks, regulatory risk is a primary concern for both investors and firms (Stroebel & Wurgler, 2021). In the context of biodiversity risk, regulatory risk becomes particularly salient. As governments and international organizations increasingly strengthen legislation and regulatory frameworks for the protection of natural capital, firms are facing mounting pressure to preserve biodiversity. Consequently, banks and investors are becoming increasingly sensitive to potential regulatory tightening in this domain. This study provides empirical evidence that banks are more responsive to biodiversity transition risk arising from stricter regulatory policies than to physical risk. This finding highlights banks' differentiated sensitivity to various types of biodiversity risk and contributes to the growing strand of empirical literature on the heterogeneity of environmental risk pricing in lending markets, as documented by Alessi, Di Girolamo, Pagano, and Giudici (2024), Bolton and Kacperczyk (2023), and Jawadi, Rozin, and Cheffou (2024).

The structure of this paper is as follows: Section 2 reviews the relevant literature, focusing on the effects of biodiversity risk on firms and their pricing in financial markets. Section 3 outlines the data, key variables, and models used in the analysis. Section 4 presents the baseline findings, robustness checks, and an investigation into the mechanisms driving the formation of the biodiversity risk premium. Section 5 explores differences between biodiversity physical and transition risks in loan pricing and discusses how banks address biodiversity risk through non-price loan terms. Finally, Section 6 concludes with a summary and policy implications.

2. Literature review and research hypothesis

2.1. Literature review

In recent years, the issuance of the Kunming Declaration, coupled

³ Transition risk arises from uncertainties associated with the structural adjustment of the socio-economic system toward a more sustainable development pathway, including technological innovation and shifts in market preferences (Alessi et al., 2024). Regulatory risk, while often considered a subset of transition risk, specifically refers to risks stemming from changes in policies, laws, and regulatory requirements. In the context of strengthened environmental governance, firms may face increased compliance costs or operational constraints if they fail to meet evolving regulatory standards (Stroebel & Wurgler, 2021).

with the tangible devastation and economic losses caused by biodiversity risk, has elevated biodiversity risk to a focal point among contemporary environmental risks (Cherief et al., 2025; Giglio et al., 2023; Huang, Cr ti, Jiang, & Sanin, 2024). The prior research relevant to the topic of this study can be categorized into two main areas: the impact of biodiversity risk on firms and the pricing of biodiversity risk in financial markets.

2.1.1. The impact of biodiversity risk on firms

The continued loss of biodiversity has been widely recognized as a critical global challenge, exerting significant impacts on both macroeconomic policymaking and microeconomic activities (Cherief et al., 2025; Fu et al., 2024; Johnson et al., 2024). Existing research indicates that exposure to biodiversity risk imposes substantial negative effects on firms' operational activities and financial performance. For instance, Bach et al. (2025) found that biodiversity risk significantly hindered the operational performance of U.S. companies, not only reducing sales growth and profitability but also increasing the cost of goods sold. Adamolekun (2024) examined the combined effects of climate risk and biodiversity risk, revealing that these dual environmental risks undermined firms' financial stability and profitability, substantially increasing the likelihood of bankruptcy. Yang and Li (2025) found that biodiversity risk weakens firms' risk-taking capacity and asset turnover, thereby intensifying their long-term reliance on short-term debt. Further evidence indicates that biodiversity risk negatively affects firms' cash holdings (Ahmad & Karpuz, 2024), operational efficiency (Li et al., 2025), and institutional ownership (Wang, 2025), and increases the likelihood of stock price crashes (Bassen et al., 2024; Liang et al., 2024).

However, some scholars, adopting a non-financial perspective, have identified potential positive implications of biodiversity risk for firms. Pi, Jiao, Zhou, and Shi (2025) found that biodiversity risk reduced ESG rating divergence by enhancing ESG disclosure and increasing transparency. He, Duan, Lucey, & Hao (2025) showed that biodiversity risk could improve corporate ESG performance by promoting green innovation and strengthening external oversight. Similarly, Tian and Chen (2025) documented that firms exposed to higher biodiversity risk are more likely to invest in ecological innovation. In addition, several studies have explored related topics such as biodiversity risk and seasoned equity offerings (Liu, Li, & Krishnamurti, 2025), as well as biodiversity conservation and firm environmental disclosure (Chen, Yin, Hu, & Li, 2025).

2.1.2. The pricing of biodiversity risk in financial markets

At present, research on the pricing of biodiversity risk primarily focuses on equity markets. For example, Giglio et al. (2023) constructed industry-based biodiversity risk exposure portfolios and found that, in response to negative biodiversity-related news, the overall returns of these portfolios were positive. Based on this observation, the authors concluded that biodiversity risk is already priced into the stock market. Kalhor and Kyaw (2024), using an event study approach, found that firms actively managing biodiversity risk experienced positive abnormal stock returns following the release of the Kunming Declaration and the 2022 UN Biodiversity Conference. Conversely, firms neglecting biodiversity risk showed little to no impact. Garel et al. (2024) observed that companies with higher CBF faced negative stock price reactions shortly after the Kunming Declaration and the launch of the Taskforce on Nature-related Financial Disclosures (TNFD). However, the average investor response over the entire sample period (2019–2022) indicated that biodiversity risk was not consistently reflected in stock prices. Ma et al. (2024) found that investor attention to biodiversity significantly enhances the excess returns of biodiversity-related stock indices. El et al. (2025) found that the biodiversity risk index exhibits significant predictive power for stock market returns. Taken together, these findings suggest that biodiversity risk has been increasingly recognized by equity market participants and priced with a higher risk premium.

In practice, due to the limited upside and significant downside risk

exposure faced by debt investors, the debt market is more sensitive to a firm's potential default risks compared to the equity market (Merton, 1974). Thus, analyzing whether biodiversity risk is priced in the debt market deserves greater attention. At present, only a limited number of studies have focused on this issue, and the findings remain inconclusive. On the one hand, in the bond market, Soylemezgil and Uzmanoglu (2024), utilizing data from the U.S. firm bond market, found that for each standard deviation increase in a firm's biodiversity sensitivity, the yield spread on its long-term bonds increased by 4 %. Similarly, Cherief et al. (2025) showed that negative biodiversity events widen the bond yield spreads of firms with greater exposure to biodiversity risk in Australia and Brazil. This finding highlights the bond market's sensitivity to biodiversity risk. However, Giglio et al. (2023), using data from the U.S. municipal bond market, constructed long-short municipal bond portfolios based on biodiversity risk exposure rankings and found no evidence of a biodiversity risk premium in the bond market. On the other hand, in the loan market, Becker et al. (2025), using data from EU banks, found a positive relationship between loan pricing and borrowers' exposure to biodiversity. Similarly, Erten and Ongena (2024) documented that while biodiversity risk is associated with higher lending rates, it is not the primary driver of environmental risk pricing in bank loans.

Based on the above analysis, it is evident that firms' biodiversity risk exposure has impacted both operational effectiveness and financial performance. However, the spillover of biodiversity risk in financial markets, whether in equity or debt markets, remains a subject of debate. Banks, as key intermediaries in the financial system, significantly influence capital flows and resource allocation based on their assessment of environmental risks. However, existing research on whether banks can effectively identify and price biodiversity-related risks remains in its early stages. The findings are still inconclusive, and there is limited investigation into the potential drivers of biodiversity risk pricing in bank lending markets. This study uses the Kunming Declaration as an external shock to analyze the impact of biodiversity risk on syndicated loan spreads and explores the main factors affecting its pricing.

2.2. Research hypothesis

The Kunming Declaration represents a pivotal policy initiative in the global effort to protect biodiversity (Nicholson et al., 2024), attracting attention from numerous scholars (Garel et al., 2024; Kalhor & Kyaw, 2024). The declaration underscores the urgency of biodiversity conservation and seeks to mobilize collective efforts to establish broad cooperative mechanisms. The declaration covers several critical areas, including enhancing ecosystem protection and restoration, promoting sustainable use, increasing public awareness, and fostering participation, aiming to construct a global collaborative framework for biodiversity conservation (Swinfield, Shrikanth, Bull, Madhavapeddy, & Zu Ermgassen, 2024). The declaration embodies the international community's consensus and commitment to biodiversity preservation, potentially accompanied by a series of policies, regulations, or regulatory measures. These initiatives not only promote global biodiversity conservation but also intensify the physical and transitional risks related to biodiversity that businesses face (Garel et al., 2024).

From the perspective of physical risk, the Kunming Declaration places a strong emphasis on biodiversity conservation, which may trigger restrictions on the overexploitation of natural resources. In the short term, such restrictions could increase the costs of acquiring critical raw materials or disrupt supply chains (Coqueret et al., 2025). For instance, industries such as mining, agriculture, and fisheries may encounter resource supply constraints due to conservation policies, including prohibitions on fishing and restrictions on logging. This directly escalates the physical risk associated with their production activities, as these sectors depend on a consistent supply of natural resources to sustain operations. Furthermore, many firms' supply chains are deeply intertwined with global biodiversity hotspots, which could

undergo significant transformations due to the implementation of biodiversity protection (Green et al., 2019). Recent studies suggest that companies exposed to significant biodiversity risk may be compelled to maintain substantial capital reserves as a precautionary measure. This requirement forces a reallocation of capital from other productive activities, thereby adversely affecting the overall efficiency of the firm (Li et al., 2025).

From the perspective of transition risk, the Kunming Declaration has spurred many countries to implement stricter ecological protection policies and regulations. These measures require firms to reduce the negative effects of their operations on biodiversity. This, in turn, necessitates firms to bear additional compliance costs (Coqueret et al., 2025), such as investing in environmentally friendly equipment, conducting environmental monitoring and management, and providing ecological compensation. Firms that fail to adapt promptly may face the risks of being fined, having their business operations restricted, or even being forced to suspend production due to failure to meet ecological protection standards, thereby increasing the pressure on their transformation (Garel et al., 2024). Secondly, the declaration reinforces global attention toward the concept of sustainable development (Cooper et al., 2023), further encouraging consumer preferences for eco-friendly products. Firms that fail to offer products meeting ecological standards may find themselves at a competitive disadvantage, leading to potential damage to their brand reputation and a decline in market share. Finally, existing research demonstrates that financial institutions and investors are increasingly prioritizing firms' environmental performance (Chen, Hasan, Lin, & Nguyen, 2021; Huang et al., 2024; Wang, Zhang, Wan, & Chen, 2023). The Kunming Declaration has further heightened the focus of financial institutions and investors on firms' biodiversity conservation, mandating that firms disclose their action plans and outcomes (Xiong, 2023). Firms failing to meet biodiversity conservation requirements may encounter difficulties in securing financing, thereby losing access to critical financial support during the transition process.

In summary, the overlapping effects of biodiversity physical and transition risks pose significant challenges to firms' operations, potentially undermining profitability and cash flow. This deterioration could exacerbate firms' credit conditions, heighten default risks, and thereby generate potential risks for financial institutions, investors, and other stakeholders. Banks, as a key source of firms' financing, often place greater emphasis on risk control and compliance management compared to other investment institutions (Kandrac & Schlusche, 2021). This focus on compliance and prudent management renders banks particularly sensitive to biodiversity-related risks in their credit activities. First, banks have accumulated extensive client data through long-term business operations. By leveraging big data analytics, banks can conduct in-depth assessments of firms' geographic locations, industry characteristics, and resource utilization (Berger et al., 2010), thereby identifying their exposure to biodiversity risk. Second, during the process of serving firms, banks typically maintain close connections with multiple industries and regulatory authorities, enabling them to grasp the policy orientations and regulatory requirements related to the environment (Wang et al., 2023). This advantage enables banks to integrate the latest policy directives with firms' operational realities, identifying risks related to firms' compliance with biodiversity regulations. Third, increasing evidence suggests that bank credit allocation affects local ecosystems (Hadji-Lazaro et al., 2024), and the resulting ecological damage may disrupt firm performance, thereby heightening banks' sensitivity to biodiversity-related risks (Becker et al., 2025). Consequently, banks are strongly motivated to internalize biodiversity-related risks within their pricing frameworks.

Based on this, the study argues that banks, leveraging their unique informational advantages, can effectively identify the potential default risks faced by firms due to increased biodiversity risk. According to risk asset pricing theory (Fama & French, 1993), when banks extend loans to firms with higher biodiversity risk, they are likely to demand higher risk premiums to compensate for potential losses. Accordingly, this research

proposes **Hypothesis 1**:

Hypothesis 1. Rising biodiversity risk, induced by the Kunming Declaration, will lead banks to claim higher loan spreads from firms.

Based on the theories of information asymmetry (Stiglitz & Weiss, 1981) and signaling (Leland & Pyle, 1977), banks gradually incorporate firms' biodiversity risk into their pricing mechanisms through an incremental process. This is primarily because biodiversity risk, as a new type of environmental risk, is currently characterized by limited availability, accuracy, and usability of relevant information (Giglio et al., 2023). Under conditions of information asymmetry, banks require progressive observation and analysis to transmit, absorb, and incorporate biodiversity information into their pricing models. In fact, as a novel form of environmental risk, the pricing of firms' biodiversity risk by banks is influenced by key driving forces or contextual factors. From the dual perspectives of firm demand and bank supply, this paper seeks to explore the role of firms' debt-paying ability and banks' green commitments in shaping the pricing of biodiversity risk.

Firms' debt-paying ability is a key indicator of their financial health and ability to meet debt obligations, directly reflecting their resilience to external shocks (Altman, 1968). Banks adjust risk premiums in their lending practices based on changes in firms' default probabilities (Duffie, Saita, & Wang, 2007). At the same time, the strength of firms' debt-paying ability directly influences the volatility of their default probabilities in response to environmental risks. On the one hand, strong debt-paying ability indicates that firms can withstand a certain level of financial stress and buffer against potential environmental risks (Guo & Zhang, 2023). For banks, even if firms incur increased costs due to biodiversity risk, such as ecological restoration expenses, firms' robust debt-paying ability can reduce the likelihood of loan defaults. On the other hand, firms with strong debt-paying ability are typically more flexible in resource allocation, enabling them to allocate more funds to manage environmental risks, such as improving production processes, adopting sustainable technologies, or implementing biodiversity conservation measures (Zhang, Bai, & Wang, 2024). In contrast, firms with weaker debt-paying ability are more likely to neglect biodiversity protection obligations due to financial constraints, thereby facing higher compliance risks and reputational pressure, which ultimately increases default risks. Accordingly, this study proposes **Hypothesis 2**:

Hypothesis 2. Firms with weaker debt-paying ability are more vulnerable, which increases banks' sensitivity to risk and results in a higher biodiversity risk premium.

As environmental protection regulations are strengthened globally, many banks are increasingly focusing on environmental impact and social responsibility when providing financial services to firms. This focus, which is aimed at building competitive advantage, enhancing brand reputation, complying with regulatory and policy requirements, and reducing long-term financial risks, is often manifested through green commitments. These commitments entail declarations of prioritizing ecological conservation and supporting sustainable development. Banks that make such commitments are commonly referred to as green banks (Delis, Greiff, Iosifidi, & Ongena, 2024; Ehlers, Packer, & De Greiff, 2022). On the one hand, green commitments drive banks to place greater emphasis on firms' biodiversity conservation performance. Through their pricing mechanisms, banks incentivize firms to minimize ecological destruction. On the other hand, a green commitment reflects a bank's focus on social image and brand reputation. If a green bank is found to support environmentally harmful companies, it may face significant public scrutiny (Bose, Khan, & Monem, 2021). By rigorously reflecting biodiversity risk in their pricing strategies, green banks can avoid excessive association with high CBF firms while demonstrating responsible behavior in environmental conservation. This approach helps to preserve green banks' public image and market reputation.

Furthermore, an increase in green banks participating in syndicated loan decision-making processes means that these banks can monitor

each other and mutually promote stricter environmental risk management (Delis et al., 2024). This consensus enhances the importance attached to biodiversity risk. Therefore, we conclude that green banks are more likely to charge higher risk premiums to firms with significant biodiversity risk exposure to mitigate potential default and reputational concerns. Moreover, this requirement for risk premiums is likely to increase as the number of green banks in the syndicate rises. Accordingly, this study proposes **Hypothesis 3**:

Hypothesis 3. Green banks will enhance the pricing of biodiversity risk, and this positive effect will strengthen as the syndicate's level of greenness increases.

3. Data and model

3.1. Data

3.1.1. Loan data

This study obtains syndicated loan data from the DealScan database. Each entry in the DealScan database includes details such as loan credits, loan terms, and covenants. Our study utilizes loan data from DealScan spanning the years 2013 to 2023. To assess the cost of bank loans for firms, we rely on two variables: margin (the interest rate charged above the London Interbank Offered Rate) and all-in-spread-drawn (includes margin along with additional fees imposed by the lender) (Acharya, Berger, & Roman, 2018; Ehlers et al., 2022). In this study, we designate these two variables as Margin and All_pricing,⁴ respectively. We extract the primary SIC codes and relevant financial data of the borrowers from the COMPUSTAT database. Each loan entry in DealScan is then merged with the financial information of borrowers from COMPUSTAT. The matching process employs the GVKEY identifier and uses the linking file developed by Chava and Roberts (2008). However, this linking file only extends to April 2018, limiting the matched dataset to loans issued before 2018. Since the policy under study was implemented in October 2021, it is necessary to include loan data beyond 2018. To address this limitation, we employ a combination of fuzzy matching⁵ and manual verification, building upon the methodology of Chava and Roberts (2008). This approach ensures the accurate and consistent matching of loan data with financial data for the period after 2018.

3.1.2. Biodiversity risk data

The dual materiality concept highlights that firms not only impact biodiversity but are also affected by it in their operations (Adamolekun, 2024; Giglio et al., 2023). Coqueret et al. (2025) applied tools developed by the TNFD and the Science-Based Targets Network (SBTN) to measure firms' exposure to physical risk. They also referenced the CBF concept introduced by Garel et al. (2024) to assess transition risk. Therefore, this paper follows the methodology of Coqueret et al. (2025) to comprehensively assess firms' biodiversity risk exposure through this evaluation framework. Based on the study by Coqueret et al. (2025), the analysis of industry-level biodiversity risk was divided into three tiers, ranging from industries with minimal dependence and impact on biodiversity to those with the highest dependence and impact. Industries classified as highly exposed to biodiversity double materiality are those with the greatest dependence and impact on biodiversity. Subsequently, we match the industry-level biodiversity scores obtained with the SIC industry codes from the COMPUSTAT database, constructing a dataset that integrates Dealscan loan data, COMPUSTAT financial data, and

⁴ All results related to All_pricing can be found in the appendix.

⁵ The initial matching was performed using the fuzzy merge algorithm in STATA, utilizing the loan firm's name and year. Given that the fuzzy merge employs a bigram string comparator and is not infallible, we manually verified all matching results.

industry-level biodiversity risk data.⁶ In this study, firms with high biodiversity risk exposure are designated as the treatment group, while firms with low biodiversity risk exposure serve as the control group. $Treat = 1$ for these highly exposed industries and $Treat = 0$ for those that are not. Using the release of the Kunming Declaration as a policy intervention point, the timeline is divided into two phases: before and after the policy implementation. $Post = 1$ for years equal to or greater than 2021 and $Post = 0$ otherwise.

Finally, to ensure the accuracy and reliability of the data, we apply the following procedures to the merged dataset: (1) Following the literature (Fard et al., 2020; Hrazdil, Anginer, Li, & Zhang, 2023; Javadi & Masum, 2021), we exclude loans associated with all financial services firms and utilities, specifically those classified under SIC codes 6000–6999 and 4900–4999. (2) The risk pricing of small-scale loan transactions tends to be more fragmented, making it challenging to derive consistent research conclusions. Therefore, our focus is on transactions involving loan amounts of at least \$100 million (Ehlers et al., 2022). (3) We excluded loans with a duration of less than four years because biodiversity risk, similar to climate risk (Huang, Kerstein, Wang, & Wu, 2022), is considered long-term.⁷ Thus, we assume that banks lack adequate time to observe and evaluate the effects of biodiversity risk on short-term loans. (4) Observations with missing key variables were excluded, and all continuous variables were winsorized at the 1 % level to reduce the impact of outliers. Finally, our regression sample consists of 677 firms and 31,018 bank-loan-firm observations, each containing comprehensive information on both borrower and lender characteristics.

3.2. Descriptive statistics

As reported in Table 1, on average, firms finance around 36.29 % of their assets through debt. The mean loan spread is 203 basis points, with a standard deviation of 103 basis points. The typical loan amount is approximately \$1.437 billion, and the average maturity period is 61 months. Each loan involves approximately 14 lenders. In our sample, about 54 % of loans are secured, and 48 % include financial covenants. These findings align with Javadi and Masum (2021), supporting the reliability of our data. The mean value of $Treat$ is 0.2523, indicating that firms with high biodiversity risk represent 25.23 % of the sample. The average value of $Post$ is 0.076, showing that 7.6 % of the observations occurred after the Kunming Declaration. Summary statistics for firm-level control variables closely match those reported by Hrazdil et al. (2023). For example, the mean ROA is 9.72 %, and the average proportion of tangible assets is approximately 25.39 %. Detailed information on the sources and measurement of all key variables is provided in the appendix Table A1. Moreover, we provide descriptive statistics on loan maturity and borrower country for the sample used in the Appendix. Fig. A1 presents the distribution of loan maturities. Fig. A2 reports the distribution of borrowers by country, while Fig. A3 illustrates the distribution of borrowers by country type.

⁶ This study uses industry-level biodiversity risk indicators, which are matched to the firm level.

⁷ This study focuses on long-term loans. On one hand, biodiversity risk represents a long-term environmental threat whose impact on firms often takes time to materialize. For instance, issues such as ecosystem degradation, extinction of key species, and disruptions in the supply of natural resources may not cause immediate financial distress but can gradually accumulate into systemic operational risks over several years (Huang et al., 2024). On the other hand, due to the extended risk exposure window associated with long-term lending, banks have stronger incentives to assess and internalize environmental risks. According to risk pricing theory (Fama & French, 1993), the longer the loan maturity, the more banks must account for future uncertainties—particularly non-traditional risk factors that could affect a firm's long-term repayment capacity, such as supply chain disruptions, regulatory pressures, or reputational damage resulting from biodiversity loss.

3.3. Model

To determine whether biodiversity risk is considered by banks, we employ a loan-level DiD model for estimation. Refer to the research design of Javadi and Masum (2021), our baseline regression takes the following form:

$$\text{Margin}_{ijt} = \alpha_0 + \beta \text{DID}_{it} + \alpha_1 \text{Treat}_i + \alpha_2 \text{Post}_t + \alpha_3 X_{it-1} + \alpha_4 Z_{ijt} + \gamma_i + \mu_t + \varepsilon_{ijt} \quad (1)$$

In the equation, i represents the borrower, j represents the lender, and t represents the year in which the loan is issued. DID_{it} is formed by the interaction of the predefined Treat_i and Post_t variables. Treat_i is a dummy variable measuring whether a firm is highly exposed to biodiversity risk, while Post_t is a dummy variable assessing whether the year of loan origination is 2021 or later. The dependent variable (Margin_{ijt}) includes Margin and AllPricing .⁸ A significantly positive β indicates that banks have considered biodiversity risk in loan pricing. Following the literature (e.g., Delis et al., 2024; Ehlers et al., 2022; Hrazdil et al., 2023; Javadi & Masum, 2021), we incorporate various control variables that could affect the cost of bank loans, including borrower, loan, and lender characteristics.

The first set of control variables (X_{it-1}) accounts for borrower characteristics, including Total Assets (Asset), Sales (Sale), Return on Assets (ROA), Leverage, Cash, Tobin Q, Operating Expenses (OpEx), Property, Plant, and Equipment (Ppe), Interest Coverage (IntCoverage), and Age. To mitigate endogeneity biases related to simultaneity and omitted variables, all borrower characteristic controls are lagged. The second set of control variables (Z_{ijt}) captures loan and lender characteristics, including Loan Value (Loan_Value), Loan Maturity (Loan_Maturity), the presence of financial covenants (Covenant), whether the loan is secured (Secured), whether it is a revolving loan (Revolver), the loan currency (Currency), the general purpose of the loan (General), the number of lenders (Lenders), the number of lead arrangers (Lead_Arrangers), and the number of lead managers (Lead_Managers). All regression models control for year (μ_t) and industry (γ_i) fixed effects, with standard errors clustered at the lender level.

4. Empirical results

4.1. Baseline regression

This study focuses on long-term loans, implicitly assuming that the biodiversity risk premium in loan pricing is primarily driven by long-term loans. To test this assumption, regressions are conducted using the full sample—including loans with maturities shorter than four years—with the results presented in Column (1) of Table 2. Further, we partition the sample based on loan maturity into a long-term loan group (maturity ≥ 4 years) and a short-term loan group (maturity < 4 years), and conduct separate regressions for each subgroup. The results are reported in Columns (2) and (3), respectively. In Column (2), the DID coefficient is statistically insignificant, while in Column (3), it is significantly positive at the 1 % confidence level. This indicates that the biodiversity risk premium required by banks is primarily present in the long-term loan group, whereas the relationship is not significant in the short-term loan group. In other words, the pricing of biodiversity risk is primarily driven by long-term lending. In Column (1), the DID coefficient is also significantly positive at the 5 % confidence level, though its magnitude is lower than that in Column (3), which is consistent with expectations. Given the shorter maturity, short-term loans are less likely to be affected by the materialization of biodiversity risk, leading to

⁸ The main conclusions of this paper, based on using Margin as the dependent variable, remain robust when AllPricing is used as the dependent variable instead. Please refer to the detailed results provided in the appendix.

Table 1
Variable descriptive statistics.

Variables	Obs	Mean	SD	Min	Median	Max
Margin	31,018	202.6785	103.4728	45.0000	175.0000	625.0000
All_Pricing	31,018	203.9156	102.7233	45.0000	175.0000	625.0000
Treat	31,018	0.2523	0.4343	0.0000	0.0000	1.0000
Post	31,018	0.0760	0.2650	0.0000	0.0000	1.0000
Loan_Value	31,018	7.2710	1.0027	4.8283	7.3132	9.7700
Loan_Maturity	31,018	4.1171	0.1274	3.8712	4.0943	4.4308
Covenant	31,018	0.4868	0.4998	0.0000	0.0000	1.0000
Secured	31,018	0.5427	0.4982	0.0000	1.0000	1.0000
Revolver	31,018	0.7904	0.4070	0.0000	1.0000	1.0000
Currency	31,018	0.9742	0.1584	0.0000	1.0000	1.0000
General	31,018	0.7209	0.4485	0.0000	1.0000	1.0000
Asset	31,018	8.3493	1.3298	4.3762	8.3263	12.2162
Sale	31,018	7.9927	1.3079	3.3270	7.9444	11.8976
ROA	31,018	0.0972	0.0578	0.0002	0.0837	0.3156
Leverage	31,018	0.3629	0.2030	0.0000	0.3411	0.9973
Cash	31,018	0.1055	0.1094	0.0013	0.0726	0.7311
Tobin_Q	31,018	2.0195	1.2951	0.6847	1.6435	9.2151
OpEx	31,018	0.7528	0.6534	0.0362	0.5589	3.5072
Ppe	31,018	0.2539	0.2297	0.0013	0.1652	0.9031
IntCoverage	31,018	1.8462	1.2564	-1.4843	1.7550	5.7273
Age	31,018	2.6392	0.7800	0.0000	2.8904	3.8067
Lenders	31,018	13.6388	8.2031	1.0000	12.0000	51.0000
Lead_Arrangers	31,018	5.8825	4.0783	0.0000	5.0000	29.0000
Lead_Managers	31,018	0.0190	0.3213	0.0000	0.0000	8.0000

weaker sensitivity of loan pricing to such risks.⁹

Based on the results in Column (3) of Table 2, holding other factors constant, firms facing higher biodiversity risk experienced a 14.373 % (29.131/202.6785) increase in loan spread relative to those with lower biodiversity risk following the release of the Kunming Declaration. Overall, the baseline regression results provide empirical support for Hypothesis 1. The control variables also align with our expectations (Javadi & Masum, 2021). The coefficient for *Loan_Maturity* is positive, suggesting that longer loan maturities are associated with greater uncertainty in future repayment cash flows, leading banks to demand higher risk compensation. The coefficient for *Covenant* is significantly negative, indicating that stricter debt covenants can mitigate banks' risk perception, thereby reducing the cost of lending. The coefficients for *Asset* and *Age* are significantly negative. Firms that have been in operation for a long time and have entered the maturity stage are perceived as more financially reliable, making them more likely to secure syndicated loans. The coefficient for *Leverage* is significantly positive. A high financial leverage ratio indicates a high level of company debt, which typically entails greater default risk. As a result, banks require higher returns to offset this risk, leading to an increased loan spread. The presence of more lead banks can reduce information asymmetry, and enhance monitoring and governance, thereby better controlling risks and reducing the required loan spread.

4.2. Parallel trend test

Accurate identification of policy effects using the DiD approach relies on the assumption that the treatment and control groups exhibit parallel trends before the implementation of the policy. Consistent with most

studies (Beck, Levine, & Levkov, 2010; Hrazdil et al., 2023), we use 2021 as the baseline period, which marks the introduction of the policy, to construct the model.

$$\text{margin}_{ijt} = \alpha_0 + \sum_{t=-4, t \neq -1}^{t=2} \beta_t \times D_t \times \text{Treat}_i + \alpha_1 X_{it-1} + \alpha_2 Z_{ijt} + \gamma_i + \mu_t + \varepsilon_{ijt} \quad (2)$$

In this model, D_t is designated as a dummy variable. For positive t values, D_t indicates the years after the Kunming Declaration, whereas for negative t values, D_t signifies the years before the declaration's implementation. The year before the declaration (D_{-1}) enactment is omitted from the regression as a control group. The other variables are defined as per eq. (1). The main coefficient β_t captures the dynamic impacts of the Kunming Declaration. We expect to see that β_{-4} ¹⁰, β_{-3} , β_{-2} are not significantly different from zero, while β_1 , β_2 should be significantly positive. In line with existing literature practices, we present the coefficients β_t derived from eq. (2) graphically in Fig. 1, which presents the estimated coefficients along with their 95 % confidence intervals for the parallel trends test. The figure indicates that before the Kunming Declaration, the coefficients for the pre-declaration period were not significant, confirming that the treatment and control groups met the parallel trends assumption before the policy's implementation. Additionally, after the Kunming Declaration, syndicated loan spreads for the treatment group exhibited a significant upward trend. In summary, our results indicate that the parallel trends test is consistent with our expectations.

4.3. Sensitivity analysis of parallel trends

In recent years, an increasing number of scholars have argued that parallel trend tests are neither sufficient nor necessary to ensure the validity of DiD analyses (Roth, Sant'Anna, Bilinski, & Poe, 2023). Roth et al. (2023) introduced a method to evaluate the impact of parallel trend assumption violations on point estimates and confidence intervals in event studies. This approach addresses the limitations of traditional parallel trend tests. Two main methods are used: (i) relative deviation

⁹ This study further investigates how the relative length of loan maturity influences the pricing of biodiversity risk, excluding short-term loans from the analysis. Following Badoer and James (2016), this study defines loans with maturities of five years or more as long-term, and those with maturities between four and five years as medium-to-short term. The results are presented in Table A4. The empirical results indicate that the relative maturity of loans has a relatively minor impact on biodiversity risk premiums. This finding mitigates the interference of loan maturity variations in biodiversity risk pricing. The analysis that follows focuses mainly on how biodiversity risk is reflected in the pricing of long-term loans.

¹⁰ Considering the extended period prior to the policy implementation, this study employs a truncation method where years more than four years before the Kunming Declaration are treated as four years.

Table 2
Baseline model regression results.

	(1)	(2)	(3)
	Full sample	Loan term < 4 years	Loan term $\geq$ 4 years
	Margin	Margin	Margin
<i>DID</i>	14.341** (5.653)	2.061 (11.932)	29.131*** (6.560)
<i>Treat</i>	-25.520* (15.247)	97.821** (44.939)	-58.126*** (13.019)
<i>Post</i>	25.669*** (6.274)	-54.979 (45.409)	24.369*** (6.754)
Loan Characteristic			
<i>Loan_Value</i>	-8.434*** (1.569)	-18.958*** (2.542)	-2.627* (1.556)
<i>Loan_Maturity</i>	-9.880*** (2.046)	-29.315*** (3.685)	114.698*** (7.319)
<i>Covenant</i>	-24.207*** (2.470)	-44.924*** (5.571)	-21.671*** (2.151)
<i>Secured</i>	62.124*** (3.629)	27.434*** (5.963)	60.958*** (3.717)
<i>Revolver</i>	-12.892*** (2.543)	-8.507** (4.187)	-17.109*** (2.808)
<i>Currency</i>	12.402 (7.620)	40.280*** (11.504)	7.763 (8.658)
<i>General</i>	-41.822*** (2.548)	-26.293*** (4.704)	-37.181*** (1.858)
Borrower Characteristic			
<i>Asset_{t-1}</i>	-8.683** (4.215)	12.494 (13.334)	-17.296*** (3.713)
<i>Sale_{t-1}</i>	1.178 (4.236)	-25.991** (11.905)	10.199** (4.219)
<i>ROA_{t-1}</i>	72.055 (57.510)	328.525*** (109.450)	11.915 (53.964)
<i>Leverage_{t-1}</i>	38.593*** (9.551)	9.295 (24.287)	38.288*** (9.163)
<i>Cash_{t-1}</i>	-37.595*** (11.779)	-125.279*** (34.497)	-2.847 (10.196)
<i>Tobin_Q_{t-1}</i>	0.530 (3.039)	-12.377*** (4.510)	2.763 (2.912)
<i>OpEx_{t-1}</i>	-1.837 (4.600)	10.264 (10.345)	-6.984 (4.600)
<i>Ppe_{t-1}</i>	25.650* (15.177)	55.848 (38.452)	27.579** (12.186)
<i>IntCoverage_{t-1}</i>	-10.133*** (2.653)	-17.606*** (5.213)	-9.982*** (2.622)
<i>Age_{t-1}</i>	-9.035*** (1.576)	-15.034*** (4.289)	-6.873*** (1.371)
Lender Characteristic			
<i>Lenders</i>	-2.238*** (0.169)	-0.801*** (0.300)	-2.225*** (0.161)
<i>Lead_Arrangers</i>	-1.424*** (0.406)	-0.219 (0.555)	-2.254*** (0.467)
<i>Lead_Managers</i>	0.880 (1.219)	- -	0.680 (1.369)
<i>_cons</i>	449.134*** (18.405)	590.073*** (45.240)	-93.692*** (33.330)
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	39,474	8456	31,018
<i>R²</i>	0.492	0.522	0.562

Note: ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. Standard errors in parentheses. The same notation applies to the remaining tables.

constraints and (ii) smoothness constraints. This study employs both methods to perform sensitivity checks on the parallel trends assumption and conducts a sensitivity analysis on the policy effects one year after implementation.¹¹

The outcomes of the sensitivity analysis are illustrated in Fig. 2,

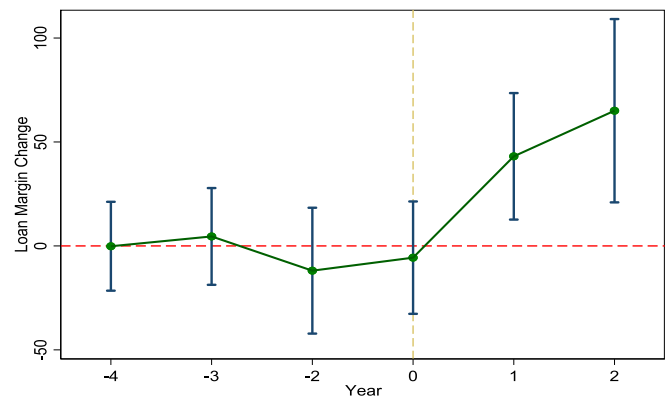


Fig. 1. Resulting of parallel trend test.

which presents robust 90 % confidence intervals for the treatment effect under the condition of partially satisfied parallel trends assumption. The vertical axis (90 % Robust CI) displays these intervals, while the horizontal axis (Mbar) measures the extent of deviation from the parallel trends. The designation original at the far left of the horizontal axis refers to the Two-Way Fixed Effects (TWFE) estimate, serving as the baseline result for the average treatment effect in this analysis. As detailed in Fig. 2, within the context of relative magnitude restrictions, when Mbar is set at 0.5, the DID coefficient is significantly different from zero at the 90 % confidence level. This suggests that the point estimate and confidence intervals maintain their robustness even when an effect that breaches the parallel trends reaches half the maximal deviation observed before policy implementation. Furthermore, under the smoothness restrictions, the confidence intervals for all measured deviations exclude zero, affirming the substantial robustness of the findings.

4.4. Placebo test

To begin with, we carry out placebo tests to mitigate concerns about the possible influence of unobservable characteristics on the DiD estimates utilized for evaluating policy effects. Following J. Wang, Qiang, Liang, Huang, and Zhong (2024), we execute these placebo tests in two different approaches. Firstly, we conduct a time placebo test. By using the one to three years before the Kunming Declaration (*Pre_1-Pre_3*) as pseudo policy shock timings, we construct $DID_i = Treat \times Pre_i$ ($i = 1, 2, 3$) replaces *DID* in the benchmark model for regression. By introducing a placebo time point, if the test results show no evidence of biodiversity risk pricing around the placebo time point, it indicates that the baseline regression results are robust. Second, this study conducts a placebo test by randomly generating placebo policy implementation dates and placebo treatment groups. The interaction of these two variables is used as the *DID* in the baseline model. This process is repeated 500 times to produce 500 placebo treatment effect estimates and their corresponding *p*-values. If the placebo effect estimates are concentrated around zero and the *p*-values exceed 0.1, it provides further evidence supporting the robustness of the baseline regression results.

As shown in Table 3 and Fig. 3, advancing the release date of the Kunming Declaration makes the policy effect insignificant. The estimated coefficients for 500 randomly generated placebo policy variables are near zero and follow a normal distribution. Most of these coefficients have *p*-values above 0.1, contrasting sharply with the baseline regression coefficient of 28.462 from the main analysis. These findings reinforce the reliability of the DiD method used in this study and highlight the robustness of the results.

4.5. Robustness tests

Alongside the primary robustness checks outlined earlier, we

¹¹ Please refer to the appendix for detailed results.

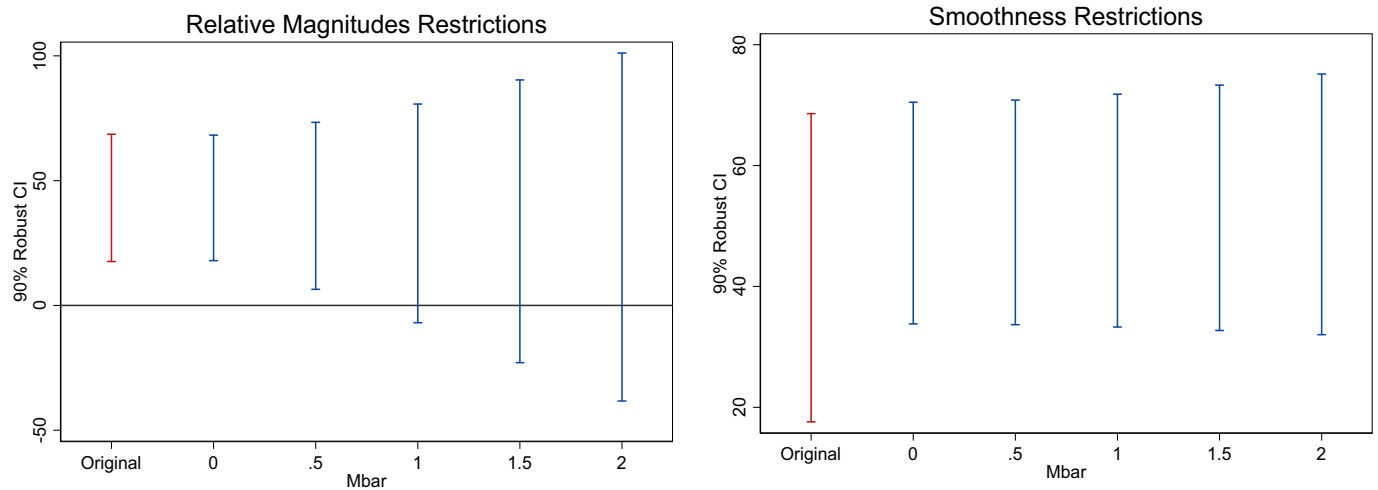


Fig. 2. Parallel trend hypothesis sensitivity test.

Table 3
Results of time placebo.

	(1)	(2)	(3)
	advanced by one year	advanced by two year	advanced by three years
	Margin	Margin	Margin
<i>DID_1</i>	3.462 (6.847)		
<i>DID_2</i>		-1.966 (6.066)	
<i>DID_3</i>			2.873 (5.018)
<i>Pre_1</i>	29.233*** (7.050)		
<i>Pre_2</i>		30.377*** (6.982)	
<i>Pre_3</i>			29.511*** (6.962)
<i>Treat</i>	-56.996*** (12.957)	-56.444*** (12.866)	-57.774*** (12.635)
<i>_cons</i>	-91.343*** (33.357)	-91.225*** (33.367)	-91.021*** (33.298)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	31,018	31,018	31,018
<i>R²</i>	0.562	0.562	0.562

perform several additional tests to ensure the reliability of the baseline regression findings. These results are presented in Table 4. Specifically, the following tests are conducted: First, fixed effects are introduced to control for unobserved variables. This approach is particularly relevant due to the notable variations in operating strategies and risk preferences among syndicate banks (Javadi & Masum, 2021), syndicate bank fixed effects are included in the analysis, with the results presented in Column (1). Additionally, to account for the varying economic cycles across different countries (Reghezza, Altunbas, Marques-Ibanez, Rodriguez d'Acrici, & Spaggiari, 2022), we incorporate borrower-country fixed effects to mitigate potential biases from omitted variables, with the results presented in Column (2). Building on Fard et al. (2020), the dependent variable is log-transformed and incorporated into the regression analysis, with the results presented in Column (3). We also apply propensity score matching (PSM) to estimate treatment effects. Using nearest-neighbor matching, each treatment group participant is paired with four control group counterparts (S. Wang et al., 2023). To address potential confounding, all control variables are included as covariates, and regressions are conducted within the common support range. The results

are presented in Column (4). Table 4 shows that the *DID* coefficients remain significant at the 1 % confidence level. This finding strengthens the reliability and validity of the study's conclusions.

This study further tests the robustness of the results by modifying the clustering method for standard errors. Two approaches are employed to adjust the standard errors of the baseline model. Following Acharya et al. (2018), the first approach addresses potential correlations in error terms. Specifically, it considers that loans issued by the same bank, to the same borrower, or within the same industry in a given year might have correlated errors, which could bias the standard errors. To mitigate this, the clustering levels for standard errors are adjusted. The results are summarized in Table 5. In Column (1), standard errors are clustered at the bank and year levels. Column (2) adjusts clustering to the borrower and year levels, while Column (3) clusters at the SIC code and year levels. Additionally, the Jackknife method is applied to minimize estimation errors through resampling (J. Wang et al., 2024). Specifically, in Column (4), 31,018 regressions are conducted by iteratively excluding one observation from the baseline sample at each step. The results in Table 5 show that the *DID* coefficients remain significant across all specifications, with no substantial changes in the regression outcomes. These findings confirm the robustness of the baseline model and strengthen the credibility of the study's conclusions.

4.6. The role of borrowers' debt-paying ability

Firms with strong debt-paying ability are better positioned to mitigate financial pressures caused by environmental deterioration and to absorb the compliance costs imposed by external regulatory policies. Such firms also exhibit greater flexibility in resource allocation and are more proactive in addressing heightened biodiversity risk (Zhang et al., 2024). In contrast, firms with weak debt-paying ability face significant difficulties in securing long-term financing. Limited financial resources make them more likely to overlook environmental obligations, further amplifying their operational vulnerabilities (Fard et al., 2020). Following Dothan (2010) and Fard et al. (2020), this study uses the interest coverage ratio (*IntCoverage*) and Altman *Z-score* to assess firms' debt-paying ability. The *IntCoverage* is defined as the natural logarithm of the ratio of earnings before interest and taxes (EBIT) to interest expenses. A higher *IntCoverage* indicates that a firm's earnings from core operations are more effective in covering interest payment obligations,

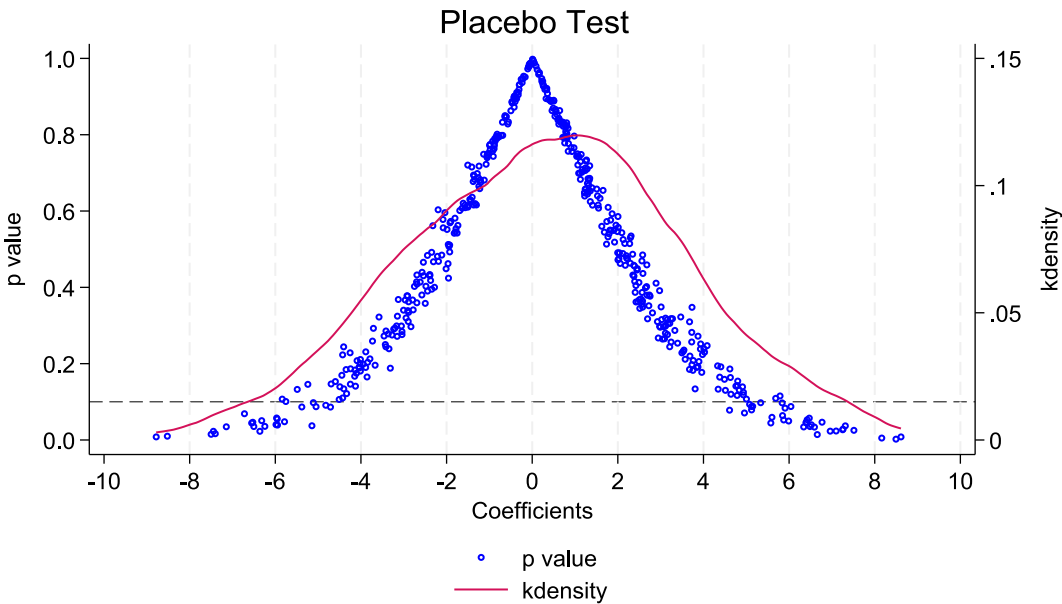


Fig. 3. Results of Placebo test.

Table 4
Results of other robustness tests.

	(1)	(2)	(3)	(4)
	Bank_syndicate Fixed Effect	Borrower_country Fixed Effect	Ln(margin)	PSM
	Margin	Margin	Ln_Margin	Margin
<i>DID</i>	19.003*** (5.574)	15.550*** (5.664)	0.135*** (0.027)	26.665*** (6.822)
<i>Treat</i>	−80.585*** (12.705)	−58.459*** (9.422)	−0.344*** (0.035)	−58.168*** (13.325)
<i>Post</i>	25.778*** (5.006)	32.779*** (4.442)	0.087*** (0.030)	11.891 (8.447)
<i>_cons</i>	−112.495*** (29.892)	−151.607*** (29.555)	4.134*** (0.127)	−99.337*** (37.033)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>Bank_Syndicate</i>	YES	YES	NO	NO
<i>Borrower_Country</i>	NO	YES	NO	NO
<i>N</i>	31,018	31,018	31,018	27,482
<i>R²</i>	0.651	0.669	0.645	0.576

Table 5
Results of different clustering of errors test.

	(1)	(2)	(3)	(4)
	Cluster by Bank & Year	Cluster by Borrower & Year	Cluster by SIC Code & Year	Jackknife
	Margin	Margin	Margin	Margin
<i>DID</i>	29.131*** (6.405)	29.131* (15.944)	29.131* (16.271)	29.131*** (6.796)
<i>Treat</i>	−58.126*** (13.804)	−58.126* (32.396)	−58.126* (33.670)	−58.126*** (13.795)
<i>Post</i>	24.369*** (7.042)	24.369 (17.192)	24.369 (17.580)	24.369*** (7.001)
<i>_cons</i>	−93.692*** (29.369)	−93.692 (80.570)	−93.692 (79.100)	−93.692*** (32.518)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>N</i>	31,018	31,018	31,018	31,018
<i>R²</i>	0.562	0.562	0.562	0.562

reflecting its short-term debt-paying ability. The Altman Z-score,¹² developed by Altman (1968), is a composite financial indicator that evaluates a firm’s financial health and long-term debt-paying ability. It incorporates measures of liquidity, profitability, leverage, and asset utilization to provide a comprehensive assessment of a firm’s financial stability. A higher Z-score suggests stronger financial stability, lower bankruptcy risk, and greater long-term debt-paying ability. These two indicators provide a comprehensive characterization of a firm’s debt-paying ability from both short-term and long-term perspectives.

In addition, credit ratings, which serve as an important indicator of a borrower’s default probability, systematically assess the firm’s ability to service debt and its overall financial stability (Cahn, Girotti, & Salvade, 2024). Credit rating agencies form assessments of a borrower’s ability to meet future debt obligations by analyzing cash flows, capital structure,

¹² The Z-score formula used in this study is: $Z_{score} = 1.2 \cdot A + 1.4 \cdot B + 3.3 \cdot C + 0.6 \cdot D + 0.99 \cdot E$. A represents capital to total assets ((act-lct)/at), B is retained earnings to total assets (re/at), C denotes pre-tax profit to total assets (ebit/at), D is the total market value to total liabilities ((prcc_c * csho)/lt), E represents sales revenue to total assets (sale/at).

and debt-paying ratios. Therefore, a borrower's credit rating serves as a key reference for banks in assessing repayment capacity during the credit decision-making process. Following the approach of [Chen, Liao, and Kuo \(2013\)](#), we use Standard & Poor's (S&P) long-term credit ratings (*Rating*) as a proxy for firm-level credit risk. The rating variable ranges from 1 to 23, where 1 corresponds to an AAA rating and 23 indicates default. A higher rating value reflects weaker credit quality and lower debt-paying ability.

In [Table 6](#), column (1) shows that the coefficient of $DID \times IntCoverage$ is significantly negative, indicating that weaker short-term debt-paying ability increases banks' sensitivity to biodiversity risk. Similarly, column (2) shows that the coefficient of $DID \times Z_score$ is also significantly negative, suggesting that weaker long-term debt-paying ability likewise heightens banks' sensitivity to biodiversity risk. In Column (3), the coefficient on $DID \times Rating$ is significantly positive, indicating that as a firm's credit rating deteriorates, banks exhibit greater sensitivity to biodiversity risk and demand higher loan spreads. These findings suggest that banks tend to charge a higher biodiversity risk premium when pricing loans for firms with weaker debt-paying ability.

4.7. The role of green banks

In this section, we examine whether green banks can fulfill their commitment to environmental responsibility by strengthening the pricing of biodiversity risk in their lending practices. Banks that recognize the challenges of social issues and commit to addressing environmental concerns are categorized as green banks. Following [Reghezza et al. \(2022\)](#), green banks are identified based on their membership in the United Nations Environment Programme Finance Initiative (UNEP FI). UNEP FI, a collaboration between the United Nations Environment Programme (UNEP) and the global financial sector, aims to integrate environmental and social responsibility into financial decision-making. It also promotes the allocation of financial resources to environmentally sustainable projects. Membership in UNEP FI is voluntary and requires adherence to principles such as the Principles for Responsible Banking (PRB) or the Net-Zero Banking Alliance (NZBA). For banks, joining UNEP FI signifies their adherence to these guidelines and their public

Table 6
The impact of borrowers' debt-paying ability.

	(1)	(2)	(3)
	Margin	Margin	Margin
$DID \times IntCoverage$	-26.335*** (8.264)		
$DID \times Z_score$		-20.740*** (3.097)	
$DID \times Rating$			2.770** (1.310)
<i>DID</i>	94.148*** (13.333)	73.799*** (19.573)	19.596 (18.450)
<i>IntCoverage</i>	-15.364*** (1.442)		
<i>Z_score</i>		-1.526** (0.710)	
<i>Rating</i>			2.843*** (0.729)
<i>treat</i>	-44.774*** (13.668)	-58.299*** (12.521)	-78.827** (36.611)
<i>post</i>	25.918*** (7.284)	35.314*** (7.184)	36.920** (14.503)
<i>_cons</i>	-70.254** (32.470)	-160.458*** (32.993)	-127.687*** (42.540)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	30,325	28,941	15,223
<i>R</i> ²	0.583	0.583	0.583

declaration of a commitment to responsible investment. This reflects a strong emphasis on environmental and social responsibility within their operations.

Information on banks participating in the UNEP FI is manually gathered from its official website.¹³ Since its inception in 1991, over 340 banks have joined this initiative. In our sample, 80 banks are members of UNEP FI. A dummy variable, *Green_Bank*, is constructed to indicate membership. *Green_Bank* equals 1 for banks that have joined UNEP FI, and 0 for those that have not. Given that a syndicated loan often involves multiple participating banks, we further consider the scenario where at least one additional bank in a loan syndicate is green. We name this variable *Green_Syndicate*, with values ranging from 0 (no green banks among the lending banks in a loan) to 17 (a loan involving 17 green banks) ([Delis et al., 2024](#)). For robustness, we also construct the variable *Green_Part*, representing the proportion of green banks to total banks in a loan syndicate, with values ranging from 0 to 1. This variable captures the extent of green bank participation in the syndicate.

In [Table 7](#), column (1), we use the interaction term between *Green_Bank* and *DID* to examine the role of green banks. The results show that $DID \times Green_Bank$ is positive, indicating that green banks genuinely pay attention to biodiversity loss, fulfilling their responsibility to support sustainable development. They charge higher credit spreads to firms with greater biodiversity risk. In columns (2) and (3), we use the $DID \times Green_Syndicate$ and $DID \times Green_Part$ to further examine the role of green syndicates. We find that the coefficients of the interaction terms are significantly positive. As the number and proportion of green banks involved in a loan increase, greater emphasis is placed on environmental factors such as biodiversity. Moreover, green banks play a more prominent role in influencing loan pricing. To summarize, the pricing of biodiversity risk in loans is influenced by the responsible attitudes of the banks. [Hypothesis 3](#) is validated.

Table 7
The role of green banks in syndicated loans.

	(1)	(2)	(3)
	Margin	Margin	Margin
$DID \times Green_Bank$	15.191** (6.884)		
$DID \times Green_Syndicate$		7.478*** (1.442)	
$DID \times Green_Part$			1.487*** (0.439)
<i>DID</i>	28.384*** (6.684)	22.969*** (7.037)	22.341*** (7.324)
<i>Green_Bank</i>	-1.425 (2.324)		
<i>Green_Syndicate</i>		0.468 (0.636)	
<i>Green_Part</i>			-28.546* (14.779)
<i>Treat</i>	-57.968*** (13.015)	-57.337*** (12.979)	-56.121*** (12.941)
<i>Post</i>	24.330*** (6.770)	26.910*** (7.161)	23.572*** (6.991)
<i>_cons</i>	-94.107*** (33.104)	-91.318*** (34.627)	-102.174*** (33.371)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	31,018	31,018	31,018
<i>R</i> ²	0.562	0.563	0.563

¹³ <https://www.unepfi.org/>

Table 8

The difference between biodiversity Transition risk and physical risk in bank loan pricing.

	(1)	(2)	(3)	(4)
	Measure reference Coqueret et al. (2025)		Measure reference Giglio et al. (2023)	
	Margin	Margin	Margin	Margin
<i>DID_Transition</i>	35.848*** (4.248)		15.270*** (4.549)	
<i>Treat_Transition</i>	25.235** (11.480)		−16.933*** (5.469)	
<i>DID_Physical</i>		17.796*** (4.894)		5.346 (3.760)
<i>Treat_Physical</i>		−22.620** (9.451)		−5.811 (6.532)
<i>Post</i>	14.889** (7.100)	20.920*** (7.102)	25.701*** (6.967)	27.992*** (6.917)
<i>_cons</i>	−121.150*** (33.673)	−96.949*** (32.928)	−103.092*** (32.996)	−104.692*** (33.126)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>N</i>	31,018	31,018	30,960	30,960
<i>R</i> ²	0.562	0.561	0.561	0.561

5. Further analysis

5.1. Biodiversity risk: Transition risk vs. physical risk

In the above research, we adopt a double materiality approach to assess biodiversity risk by averaging both physical and transition risks. However, transition risk and physical risk are inherently different (Giglio et al., 2023). Similar to how carbon risk received increased attention following the Paris Agreement, which required higher risk premiums for borrowers with high carbon emission intensity (Ehlers et al., 2022), biodiversity transition risk should receive more focus after the Kunming Declaration, as it represents a primary biodiversity risk faced by firms. The Sakhalin-II oil and gas development project, financed by a syndicate led by the European Bank for Reconstruction and Development (EBRD), serves as a significant case in point. The project's wastewater discharges severely damaged the local aquatic ecosystem and disrupted the livelihoods of local fishermen. In response, non-governmental organizations (NGOs) filed a lawsuit against the Ministry of Natural Resources of the Russian Federation. Ultimately, the project was halted, despite being 90 % complete and having consumed billions of dollars in investment, resulting in significant losses for the lending banks. In contrast, physical risk, unlike transition risk, tends to have localized impacts rather than global effects (Bolton & Kacperczyk, 2023). Firms can mitigate biodiversity physical risk by adjusting resource acquisition strategies or reconfiguring supply chains. As a result, the long-term impact of biodiversity physical risk on firms is relatively limited, which alleviates the premium banks claim for such risks.

To analyze the differences in how biodiversity transition and physical risks are priced in bank loans and to identify which type of risk is emphasized, we follow the approach of Coqueret et al. (2025). We use industry-level CBF/Revenue and SBTN significance scores to define the variables *treat_Transition* and *Treat_Physical*.¹⁴ The values of CBF/Revenue range from −1 to 0, with larger absolute values indicating greater negative impacts on biodiversity. SBTN, on the other hand, takes positive values, with higher values reflecting a greater dependency of firm operations on biodiversity. *Treat_Transition* is assigned a value of 1 for scores below the industry median and 0 otherwise. Similarly, *Treat_Physical* is set to 1 for scores above the industry median and 0 otherwise. These variables are then interacted with *Post* to generate the interaction

terms *DID_Transition* and *DID_Physical*. Additionally, to ensure robustness, we alter the measurement of *Treat_Transition* and *Treat_Physical*. Following Giglio et al. (2023), who developed an industry-level measure of biodiversity risk based on surveys of financial scholars, professionals, and regulators. The scholar's two biodiversity risk indicators were positive, with higher values indicating a greater degree of impact on or dependency upon biodiversity. We define *Treat_Transition* and *Treat_Physical* as follows: *Treat_Transition* is assigned a value of 1 if the scores are above the industry median, and 0 otherwise. Similarly, *Treat_Physical* is set to 1 for scores above the industry median and 0 otherwise. These variables also interacted with *Post* to create the interaction terms *DID_Transition* and *DID_Physical*.

As shown in Table 8, the coefficient for *DID_Transition* in column (1) is significantly larger than that for *DID_Physical* in column (2), with a difference of approximately one factor. This indicates that while banks consider both transition and physical risks when pricing biodiversity risk, banks demand a higher biodiversity risk premium for transition risk than for physical risk. This suggests that banks are more concerned with the future compliance costs associated with stricter biodiversity policies. Additionally, the coefficient for *DID_Physical* in column (4) is not only smaller than that for *DID_Transition* in column (3) but also statistically insignificant. This may be because Giglio et al. (2023) rely more heavily on the opinions of financial experts and professionals,¹⁵ whose understanding of biodiversity physical risk might partially outpace industry practices. The forward-looking perspectives of experts often require time to be translated into banks' operational practices. This time lag could explain why biodiversity physical risk is not yet significantly reflected in loan pricing at the current stage. In the appendix, we construct *DID_Physical* and *DID_Transition* based on Giglio et al. (2023) measures. When using All Pricing as the dependent variable, *DID_Physical* is significant at the 10 % confidence level, but the coefficient remains smaller than the premium banks require for transition risk.

Overall, banks are more sensitive to biodiversity transition risk, claiming higher risk premiums. In the case of biodiversity physical risk, two factors contribute to the weaker impact on firms' loan costs. First, firms can mitigate the increased physical risk induced by the Kunming

¹⁴ The median value of CBF/Revenue is −0.08, and the median SBTN materiality score is 76.5.

¹⁵ Giglio et al. (2023) conducted a survey to assess perceptions of industries most affected by the physical and transition risks of biodiversity loss. The impact was measured based on the percentage of respondents identifying each sector as significantly affected. The participants included 48 % academics, 34 % finance professionals, and 18 % researchers from financial or public sector institutions.

Table 9

The impact of different legal systems on the pricing of biodiversity risk in loans.

	(1)	(2)
	Common law	Civil law
	Margin	Margin
<i>DID</i>	27.779*** (6.608)	110.811 (79.005)
<i>Treat</i>	−36.855*** (10.518)	−127.934*** (23.557)
<i>Post</i>	33.676*** (6.661)	−53.429** (25.205)
<i>_cons</i>	−50.195 (33.341)	248.882** (99.800)
<i>Controls</i>	YES	YES
<i>Industry</i>	YES	YES
<i>Year</i>	YES	YES
<i>N</i>	28,790	2228
<i>R</i> ²	0.551	0.943

Declaration through alternative strategies, such as resource reallocation or operational adjustments. Second, physical risk is inherently harder to quantify (e.g., the costs associated with the loss of ecosystem services). Compared to academic and financial experts, banks may lack a sufficient understanding of physical risk, leading to a weaker influence of such risks on loan pricing.

5.2. Legal system: Common law vs. civil law

The legal framework plays a foundational role in the development and functioning of a nation's financial system. Through well-defined property rights and effective enforcement mechanisms, the legal system shapes the evolution of capital markets, firms' financing structure choices, and the behavior of financial intermediaries (Ge, Kim, & Song, 2012). At the same time, it embeds key elements of financial governance, such as market efficiency, contract enforcement, and risk sharing, into the institutional architecture of the financial system (Liu, He, Men, & Sun, 2024). Against the backdrop of rising environmental risks, the type of legal system not only influences the stringency and enforcement of environmental regulations but also shapes how financial institutions and investors perceive and respond to firms' exposure to environmental risks. Ferdous, Atawnah, Yeboah, and Zhou (2024) document significant cross-country differences in investor attitudes toward climate risk under different legal traditions and provide empirical evidence that the positive association between climate risk and accounting conservatism is more pronounced in common law countries.

Currently, national legal systems are broadly classified into two categories: the common law system, represented by countries such as the United Kingdom and the United States, and the civil law system, represented by countries such as France and Germany (Graff, 2008). Common law systems emphasize judicial precedent and market self-regulation, with legal frameworks that tend to delineate behavioral boundaries rather than prescribe detailed operational guidelines (Ergungor, 2004). Under such institutional arrangements, firms are required to exhibit stronger adaptive capacity and higher standards of environmental governance when responding to biodiversity risk amid evolving market conditions and policy shifts. Consequently, financial institutions place greater emphasis on firms' governance capabilities in lending decisions and demonstrate a heightened sensitivity to biodiversity risk, demanding a higher risk premium accordingly. In contrast, civil law countries typically maintain more explicit and centralized administrative regulatory systems (Diniz-Maganini, Rasheed, & Yaşar, 2023). As a result, firms in these jurisdictions tend to rely more heavily

on external policy support when addressing environmental risks, which in turn diminishes banks' incentives to differentiate firms based on their internal environmental governance capabilities.

This study conducts a subgroup regression based on whether the borrower's home country follows a common law or civil law system. The results are presented in Table 9. Column (1) shows that the DID coefficient is significantly positive at the 1 % confidence level, indicating that in common law countries, banks demand higher loan spreads from firms exposed to greater biodiversity risk. In contrast, while the DID coefficient in Column (2) is also positive, it is not statistically significant. Overall, the results in Table 9 suggest that the pricing of biodiversity risk in syndicated loans is shaped by the legal framework, and more specifically, is driven by common law systems.

5.3. Other loan features

So far, our findings indicate that, following the Kunming Declaration, banks claim higher loan spreads from firms exposed to high biodiversity risk. However, based on the theory of credit rationing (Stiglitz & Weiss, 1981), there may be an adverse selection effect of interest rates in an imperfect information market. Specifically, higher interest rates may attract riskier borrowers to apply for loans, prompting banks to potentially deny loans to certain borrowers. As a result, banks can manage the default risks linked to borrowers' biodiversity exposure by not only adjusting loan spreads but also modifying non-price terms in loan agreements. These modifications may include reducing loan amounts, shortening maturities, increasing fees, and other measures (Fard et al., 2020; Javadi & Masum, 2021). For instance, reducing loan amounts can lower the risk associated with individual loans, while increasing upfront fees and commitment fees helps to compensate for operational and opportunity costs. Similarly, shortening loan maturities reduces the uncertainty associated with asset value declines caused by changes in market conditions. By employing such measures, banks can partially mitigate their exposure to biodiversity risk. Therefore, we hypothesize that lenders can more effectively manage biodiversity risk through alternative mechanisms.

To test our hypothesis, we follow Fard et al. (2020) and Hrazdil et al. (2023) by using eq. (1) to regress the loan amount, loan maturity, the product of loan amount and maturity, upfront fees, and annual fees as the dependent variables. $Loan_Amount \times Maturity$ is calculated as the logarithm of the product of loan amount and loan maturity, representing the total dollar-years of the loan. It reflects the bank's overall credit risk exposure during the loan term. A higher value indicates a greater financial commitment over an extended period, leading to increased credit risk (Acharya et al., 2018). *Upfront* and *Annual* represent upfront fees and annual fees, respectively. Upfront fees refer to one-time charges paid at the inception of the loan to compensate the lending institution for the time and resources invested in arranging the loan, such as assessing the borrower's creditworthiness and drafting the loan agreement. Annual fees are the charges levied by the lending institution for the commitment to provide a specified amount of credit. Even if the borrower does not utilize the entire loan amount, the lending institution charges these fees to maintain the availability of these funds.

By including these variables as dependent variables in the model (1), we aim to explore how lenders might adjust these other contractual features in response to a firm's biodiversity risk. The results presented in Table 10 are consistent with our expectations. In column (1), the *DID* coefficient is significantly negative, indicating that an increase in biodiversity risk substantially reduces the loan amount. In column (2), although the *DID* coefficient remains negative, it is not statistically significant. In column (3), the *DID* coefficient is significantly negative,

Table 10
Other contractual features of loans.

	(1)	(2)	(3)	(4)	(5)
	Loan_Value	Loan_Maturity	Loan_Amount×Maturity	Upfront	Annual
<i>DID</i>	−0.166*** (0.044)	−0.007 (0.006)	−0.173*** (0.046)	40.302*** (11.891)	4.917*** (1.147)
<i>Treat</i>	0.601*** (0.114)	−0.053*** (0.010)	0.554*** (0.114)	−71.052*** (20.699)	−21.923*** (1.483)
<i>Post</i>	−0.399*** (0.070)	0.001 (0.007)	−0.388*** (0.074)	−42.373*** (13.560)	9.588*** (2.301)
<i>_cons</i>	3.236*** (0.262)	4.130*** (0.014)	6.998*** (0.169)	105.902*** (33.948)	−10.070 (33.369)
<i>Controls</i>	YES	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES	YES
<i>N</i>	31,018	31,018	31,018	5460	14,318
<i>R</i> ²	0.683	0.275	0.679	0.539	0.545

demonstrating that an increase in biodiversity risk notably decreases the bank's exposure to credit risk, which aligns with the findings of [Javadi and Masum \(2021\)](#). Furthermore, in columns (4) and (5), the *DID* coefficients are positive, suggesting that an increase in biodiversity risk results in higher upfront and commitment fees. Banks may consider imposing stricter non-price contractual terms in syndicated loans to reduce their exposure to biodiversity risk.

6. Conclusion and policy implication

As environmental legislation becomes increasingly prevalent, investors become more conscious of sustainability, and public demands for environmental protection intensify, banks are under mounting pressure to shoulder greater social responsibilities. Beyond fulfilling their traditional economic functions, banks must also enhance their capacity to identify and manage environmental risks. This paper provides empirical evidence, based on long-term bank loan data, that banks are incorporating biodiversity risk, a novel form of environmental risk, into their lending decisions. Using global syndicated loan data from 2013 to 2023, the findings reveal that, following the Kunming Declaration, firms with high biodiversity risk exposure face higher loan spreads. This finding holds robustly across a series of robustness checks. Furthermore, this study finds that weak firms' debt-paying ability and banks' green commitments heighten banks' sensitivity to biodiversity risk exposure. Additionally, compared to biodiversity physical risk, banks are more sensitive to biodiversity transition risk, demanding higher loan spreads for transition risk. Furthermore, this paper shows that the biodiversity risk premium in syndicated loans is driven by common law systems, highlighting the role of legal institutions in shaping environmental risk pricing. Finally, beyond employing price terms, banks also address firms' biodiversity risk through various non-price measures, such as reducing loan amounts, shortening maturities, and increasing upfront fees.

These findings offer valuable insights for policy development. Firstly, this study finds that banks have started pricing biodiversity risk and internalizing the negative externalities caused by firms' heavy reliance on or impact on ecological systems. However, a considerable gap remains in meeting the Kunming Declaration's 30 × 30 initiative, which aims to protect 30 % of the world's terrestrial and marine ecosystems by 2030. Bank regulators and supervisors should work to establish a more robust green financial policy framework. On the one hand, differentiated rediscount rates should be implemented across banks to reduce the flow of funds to projects and firms that threaten ecological systems. On

the other hand, policy-driven incentive mechanisms should be introduced to guide banks in increasing financial support for projects with low ecological risk and high conservation value. This would help narrow the funding gap required to achieve the 30 × 30 initiative.

Secondly, this study finds that biodiversity risk has an adverse impact on firms' long-term financing costs. To secure lower-cost and more stable long-term funding, firms must more systematically integrate ecological conservation considerations into their business models. This is particularly important for long-maturity loans and in common law countries, where banks exhibit greater sensitivity to firms' exposure to biodiversity risk. On the one hand, firms should optimize resource allocation in their business practices by upgrading technologies and building green supply chains to reduce dependency on ecosystems, thereby better addressing both physical and transition risks related to biodiversity. On the other hand, firms should improve their biodiversity risk management strategies to increase their resilience to potential risks. This includes establishing specialized biodiversity management teams and leveraging advanced monitoring technologies to enable dynamic risk management and optimize decision-making. These efforts lay a solid foundation for achieving long-term sustainable development.

Third, this study finds that green banks demonstrate greater sensitivity and proactiveness in addressing biodiversity risk. Given that private capital still plays a limited role in biodiversity finance ([Flammer, Giroux, & Heal, 2025](#)), it is imperative for policymakers to incentivize green banks to promote the persistent engagement of green banks in mobilizing biodiversity finance. This can be achieved by implementing performance incentive schemes for green banks and providing low-interest refinancing to support their financing of environmentally friendly firms. In addition, the empirical results of this study reveal that as the increasing number of green banks in lending, sensitivity to biodiversity risk also intensifies. Therefore, policymakers should establish platforms for collaboration and knowledge exchange among green banks. Such platforms can facilitate the sharing of successful experiences in biodiversity project risk assessment and innovative financing models, thereby enhancing the banking sector's overall capacity to address biodiversity risk.

Finally, this study indicates that banks have started integrating biodiversity risk into their existing risk management systems. However, as a novel type of environmental risk, biodiversity risk lacks a standardized assessment framework and suffers from severe data scarcity. Consequently, mandating or encouraging firms to disclose biodiversity-related information becomes particularly critical. Policymakers should encourage firms to disclose the impacts and dependencies of their

operations on ecosystems. They should also urge companies to comply with standardized environmental reporting frameworks, such as those proposed by the TNFD. By providing banks with transparent and consistent environmental performance data, banks can optimize their loan pricing strategies and reduce unreasonable risk premiums caused by information asymmetry.

7. Limitations

This study offers valuable empirical evidence on how banks price biodiversity risk, but it also has certain limitations that warrant further investigation in future research. Although we adopt an international perspective, our sample is heavily concentrated in the U.S. market, which accounts for approximately 90 % of all loan observations. Loan data from other countries are relatively limited, with most individual countries representing less than 2 % of the sample. This distribution implies that the empirical findings largely reflect the characteristics of the loan market in the United States, a developed economy, and thus the generalizability of our conclusions across international contexts may be constrained. Future research could enhance external validity by incorporating a more diverse set of loan data from other regions, particularly

from emerging market economies, to assess the broader applicability of these findings.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed. There are no financial conflicts of interest to disclose.

Acknowledgments

Work on this paper benefited from the financial support of the National Natural Science Foundation of China (72374061; 72302073; 72188101), Ministry of Education Humanities and Social Science Research Youth Project (22YJC630056), the Fundamental Research Funds for the Central Universities of China (JZ2025HGTB0237) and the Anhui Provincial Natural Science Foundation (2208085UD02).

Appendix A. Appendix

A.1. Setting of variables

Table A1
Explanation of key variables.

VarName	Description
<i>Margin</i>	The variable is measured as margin in the DealScan database. Specific interest rate being charged over the base rate.
<i>All_Pricing</i>	The variable is measured as all-in spread drawn in the DealScan database. All-in spread drawn compare with margin, it has increased all types of fees charged by the lender, including commitment fees (paid on unused amount of loan commitments), utilization fees (paid on the drawn amount once a threshold has been exceeded).
<i>DID</i>	The interaction term between Treat and Post.
<i>Treat</i>	These industries are highly exposed to the dual materiality of biodiversity risks. These highly exposed industries are set as treatment groups.
<i>Post</i>	The Kunming Declaration was released in October 2021, so the Post setting is set to 1 for years greater than 2021, and 0 for others.
<i>Loan_Value</i>	Natural logarithm of the loan amount, with the loan amount measured in units of one million US dollars.
<i>Loan_Maturity</i>	Natural logarithm of the maturity expressed in months.
<i>Covenant</i>	Indication of whether or not financial covenants exist. Existence is 1, non existence is 0.
<i>Secured</i>	Indication of whether or not the tranche is secured. Existence is 1, non existence is 0.
<i>Revolver</i>	Indication of whether or not the revolving credit. The credit is reversing, the value of the credit is 1, otherwise it is 0.
<i>Currency</i>	The measurement method for loan currency is 1 for those denominated in US dollars and 0 for those denominated in other method.
<i>General</i>	If the loan purpose indicator is for general corporate purposes, capital expenditures, or working capital, the value is 1, otherwise it is 0.
<i>Asset</i>	Natural logarithm of book assets (AT in Compustat).
<i>Sale</i>	Log of sales in thousands of dollars (Compustat SALE).
<i>ROA</i>	Earnings before interest and taxes divided by total assets (Compustat EBIT / AT)
<i>Leverage</i>	Total debt, defined as debt in current liabilities plus long-term debt, divided by total assets [Compustat (DLC + DLTT)/AT].
<i>Cash</i>	Cash and short-term investments divided by total assets (Compustat CHE/AT).
<i>Tobin_Q</i>	Ratio of the sum of the year-end market capitalization and the difference between book asset value and common/ordinary equity to book asset value: $(AT - CEQ + PRCC \times CSHO) / AT$ in Compustat.
<i>OpEx</i>	Ratio of operational expenses to book assets. XOPR/AT in Compustat.
<i>Ppe</i>	Ratio of net tangible assets to book assets. PPENT/AT in Compustat.
<i>IntCoverage</i>	Natural logarithm of the ratio of EBIT to interest expenses. EBIT/XINT in Compustat
<i>Age</i>	Natural logarithm of the difference between the year of observation and the initial public offering year (using IPODATE in Compustat). If the Names file indicates a higher age, we substitute the previous measure with the Names file number.
<i>Lenders</i>	All Lender(s) who participated in the loan.
<i>Lead_Arrangers</i>	The number of lender(s) who acted as Lead Arranger on the loan.
<i>Lead_Managers</i>	The number of lender(s) who acted as a Lead Manager on the loan.

A.2. Sample data structure.

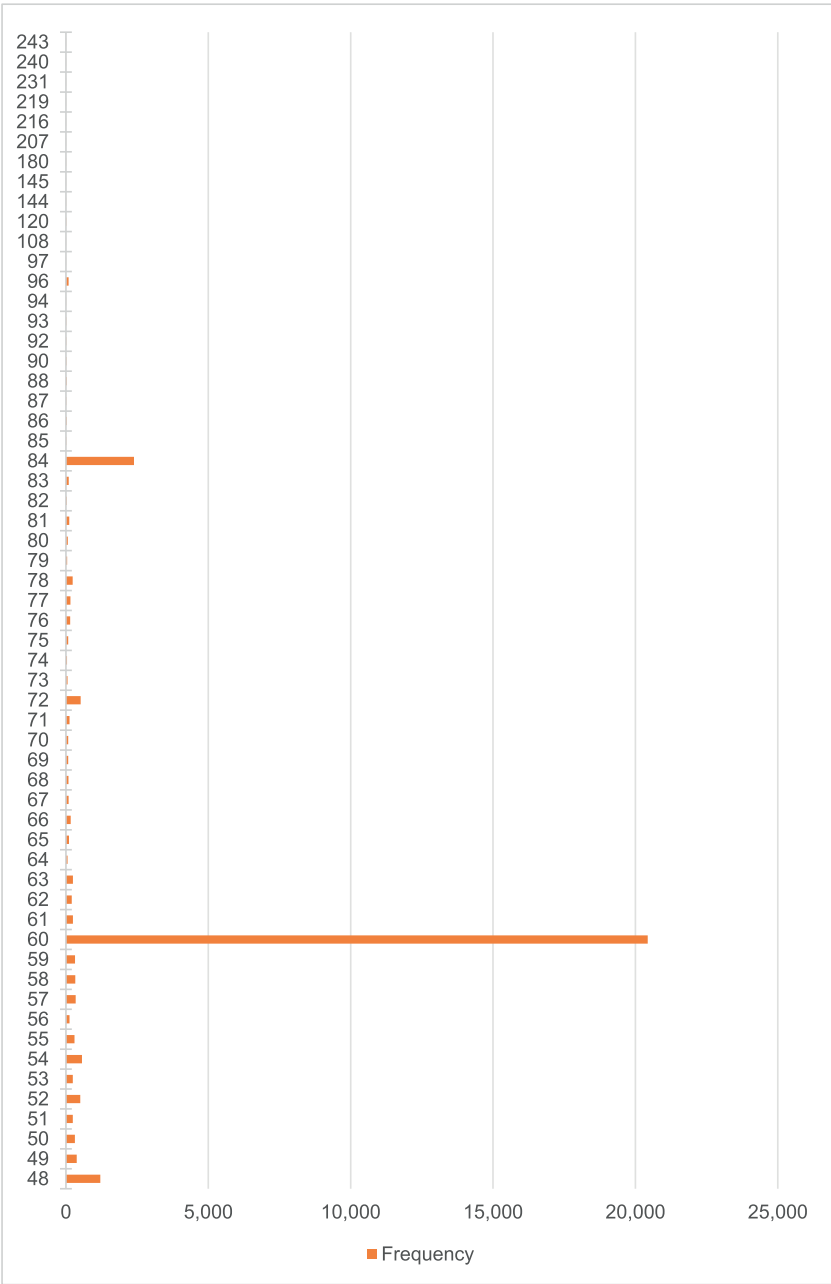


Fig. A1. Distribution of loan maturities (month).

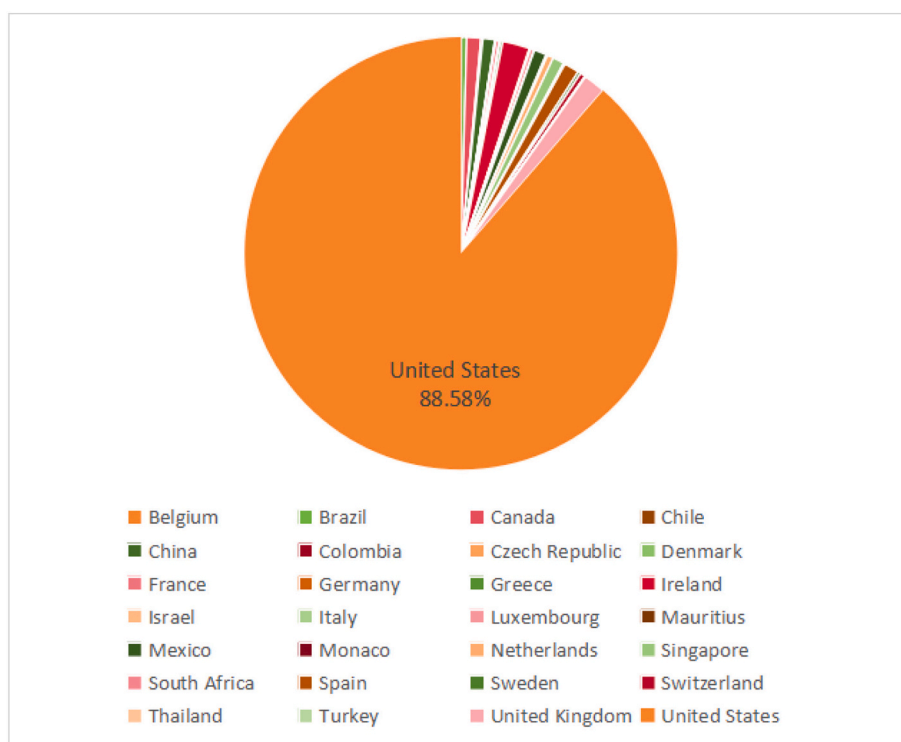


Fig. A2. Distribution of borrowers by country.

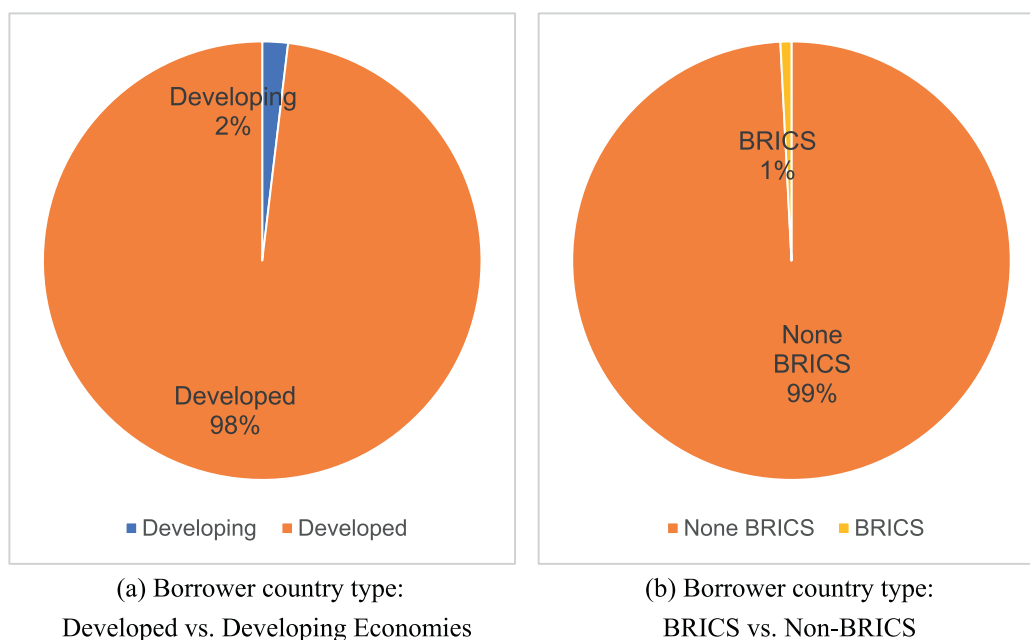


Fig. A3. Distribution of borrowers by country type.

Table A2

Double materiality industry ranking.

A	FF48Code	FF48 Industry Long Name	CBF Revenue	SBTN materiality score	CBF Ranking	SBTN Ranking	Average Ranking	Exposure
1	14	Chemicals	-0.39	223.42	5	4	4.5	High
2	39	Shipping Containers	-0.72	159.33	4	5	4.5	High
3	38	Business Supplies	-1.35	90	1	14	7.5	High
4	22	Electrical Equipment	-0.23	105.57	8	10	9	High
5	1	Agriculture	-0.84	81.29	2	18	10	High
6	2	Food Products	-0.84	81.29	3	19	11	High

(continued on next page)

Table A2 (continued)

A	FF48Code	FF48 Industry Long Name	CBF Revenue	SBTN materiality score	CBF Ranking	SBTN Ranking	Average Ranking	Exposure
7	17	Construction Materials	−0.1	154.38	18	6	12	High
8	31	Utilities	−0.06	310	27	1	14	High
9	9	Consumer Goods	−0.05	254	30	2	16	High
10	28	Non-Metallic and Industrial Metal Mining	−0.13	78.77	13	20	16.5	High
11	36	Electronic Equipment	−0.08	96.23	21	12	16.5	High
12	25	Shipbuilding, Railroad Equipment	−0.04	247	31	3	17	High
13	10	Apparel	−0.1	84.22	19	16	17.5	High
14	18	Construction	−0.07	132.08	26	9	17.5	High
15	27	Precious Metals	−0.13	78.77	14	21	17.5	High
16	29	Coal	−0.08	103.86	24	11	17.5	High
17	4	Beer & Liquor	−0.17	74	11	25	18	Medium
18	16	Textiles	−0.1	84.22	20	17	18.5	Medium
19	19	Steel Works Etc	−0.13	78.77	15	22	18.5	Medium
20	3	Candy & Soda	−0.17	74	12	26	19	Medium
21	13	Pharmaceutical Products	−0.23	49	9	31	20	Medium
22	34	Business Services	−0.07	89.38	25	15	20	Medium
23	21	Machinery	−0.06	92.97	28	13	20.5	Medium
24	30	Petroleum and Natural Gas	−0.02	136	34	7	20.5	Medium
25	37	Measuring and Control Equipment	−0.22	49	10	32	21	Medium
26	15	Rubber and Plastic Products	−0.08	76.5	22	23	22.5	Medium
27	26	Defense	−0.13	68.83	17	28	22.5	Medium
28	40	Transportation	−0.01	135.8	38	8	23	Medium
29	41	Wholesale	−0.08	76.5	23	24	23.5	Medium
30	43	Restaurants,Hotels, Motels	−0.33	16.76	6	42	24	Medium
31	5	Tobacco Products	−0.13	45	16	33	24.5	Medium
32	44	Banking	−0.24	6	7	43	25	Low
33	23	Automobiles and Trucks	−0.03	71	32	27	29.5	Low
34	6	Recreation	−0.02	62	33	29	31	Low
35	42	Retail	−0.01	50.67	37	30	33.5	Low
36	35	Computers	−0.05	26	29	39	34	Low
37	33	Personal Services	−0.02	34	35	36	35.5	Low
38	7	Entertainment	0	40.38	41	34	37.5	Low
39	12	Medical Equipment	0	34	39	37	38	Low
40	32	Communication	−0.01	26	36	40	38	Low
41	8	Printing and Publishing	0	40.38	42	35	38.5	Low
42	11	Healthcare	0	33.25	40	38	39	Low
43	45	Insurance	0	21.75	45	41	43	Low
44	47	Trading	0	6	43	44	43.5	Low
45	46	Real Estate	0	6	44	45	44.5	Low
46	20	Fabricated Products	.	.	.	.	.	Low
47	24	Aircraft	.	.	.	.	.	Low
48	48	Almost Nothing	.	.	.	.	.	Low

This table refers to [Coqueret et al. \(2025\)](#), Double materiality Industry Ranking Industries are ranked by highest exposure to biodiversity loss. Average CBF/Sales and SBTN Scores are computed at the GICS Sub-Industry level then aggregated and matched to Fama and French 48 Industry portfolios. SBTN original ratings are transformed into quantitative scores following a linear scale.

CBF expresses a firm's negative impact on biodiversity in terms of "artificialized" square kilometers of land. For example, a CBF value of −100 km² indicates a 10 % loss of original biodiversity in a 1000 km² area or a 5 % loss in a 2000 km² area. Accordingly, CBF values are always negative, with larger absolute values reflecting more severe biodiversity loss.

In this study, we use the ratio of CBF to corporate revenue (CBF/Revenue) as a standardized measure of biodiversity risk exposure. Since revenue is positive and CBF is negative, the CBF/Revenue ratio is necessarily negative. A larger absolute value of this ratio implies more extensive biodiversity degradation per unit of revenue, suggesting that banks may demand a higher risk premium when assessing environmental risks.

A.3. Regression results with explained variable changed to All-Pricing.

Table A3
Baseline model regression results.

	(1)	(2)	(3)
	Full sample	Loan term<4 years	Loan term≥ 4 years
	All_Pricing	All_Pricing	All_Pricing
<i>DID</i>	14.851*** (5.632)	3.426 (11.907)	29.665*** (6.515)
<i>Treat</i>	−25.076 (15.250)	98.201** (44.899)	−57.887*** (13.059)
<i>Post</i>	31.931*** (6.316)	−74.315* (42.641)	30.697*** (6.755)
<i>Loan Characteristic</i>			

(continued on next page)

Table A3 (continued)

	(1)	(2)	(3)
	Full sample	Loan term<4 years	Loan term≥ 4 years
	All_Pricing	All_Pricing	All_Pricing
<i>Loan_Value</i>	−8.625*** (1.586)	−19.516*** (2.618)	−2.735* (1.560)
<i>Loan_Maturity</i>	−9.434*** (2.052)	−28.435*** (3.711)	114.811*** (7.320)
<i>Covenant</i>	−23.133*** (2.489)	−44.861*** (5.575)	−20.572*** (2.147)
<i>Secured</i>	60.825*** (3.613)	25.628*** (5.941)	59.635*** (3.706)
<i>Revolver</i>	−12.498*** (2.528)	−7.997* (4.214)	−16.679*** (2.804)
<i>Currency</i>	12.092 (7.721)	39.817*** (11.696)	8.202 (8.660)
<i>General</i>	−41.368*** (2.571)	−26.673*** (4.762)	−36.775*** (1.865)
Borrower Characteristic			
<i>Asset_{t-1}</i>	−8.202* (4.211)	13.641 (13.417)	−16.926*** (3.707)
<i>Sale_{t-1}</i>	1.258 (4.245)	−27.150** (12.051)	10.456** (4.216)
<i>ROA_{t-1}</i>	69.540 (57.440)	341.930*** (109.682)	6.253 (53.805)
<i>Leverage_{t-1}</i>	39.403*** (9.581)	7.950 (24.265)	39.655*** (9.123)
<i>Cash_{t-1}</i>	−37.277*** (11.833)	−123.583*** (34.306)	−2.088 (10.234)
<i>Tobin_Q_{t-1}</i>	0.708 (3.039)	−12.396*** (4.457)	2.913 (2.907)
<i>OpEx_{t-1}</i>	−0.504 (4.604)	12.155 (10.437)	−5.845 (4.579)
<i>Ppe_{t-1}</i>	25.468* (15.236)	56.828 (38.746)	27.076** (12.198)
<i>IntCoverage_{t-1}</i>	−10.207*** (2.658)	−18.034*** (5.242)	−9.888*** (2.621)
<i>Age_{t-1}</i>	−8.852*** (1.580)	−15.167*** (4.311)	−6.652*** (1.367)
Lender Characteristic			
<i>Lenders</i>	−2.218*** (0.168)	−0.767*** (0.293)	−2.215*** (0.160)
<i>Lead_Arrangers</i>	−1.471*** (0.407)	−0.131 (0.553)	−2.304*** (0.467)
<i>Lead_Managers</i>	0.834 (1.208)	− −	0.651 (1.357)
<i>_cons</i>	443.060*** (18.536)	590.839*** (45.765)	−100.107*** (33.256)
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	39,441	8423	31,018
<i>R²</i>	0.486	0.519	0.557

Table A4

Cross-sectional analysis of loan terms.

	(1)	(2)	(3)	(4)
	4 years<Loan term<5 years	Loan term≥ 5 years	4 years<Loan term<5 years	Loan term≥ 5 years
	Margin	Margin	All_Pricing	All_Pricing
<i>DID</i>	104.638*** (16.268)	28.975*** (6.489)	109.575*** (16.545)	30.022*** (6.410)
<i>Treat</i>	17.223 (27.883)	−78.266*** (13.537)	16.634 (27.426)	−78.216*** (13.580)
<i>Post</i>	−103.924*** (21.296)	39.126*** (6.979)	−105.758*** (21.646)	45.366*** (6.939)
<i>_cons</i>	528.541*** (105.013)	−562.672*** (39.756)	577.369*** (105.589)	−568.965*** (39.701)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>N</i>	4866	26,152	4866	26,152
<i>R²</i>	0.693	0.599	0.689	0.595

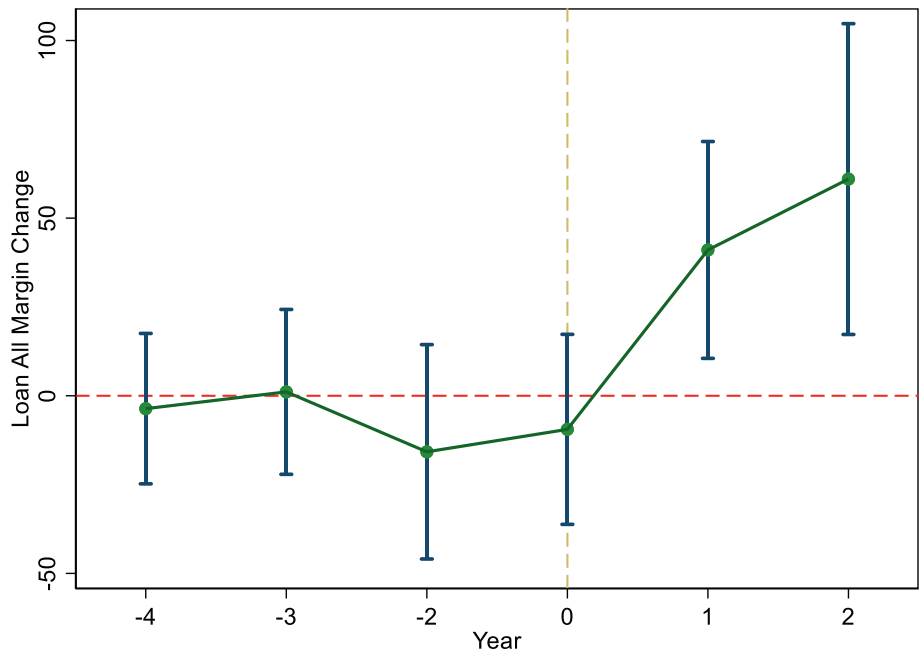


Fig. A4. Resulting of parallel trend test.

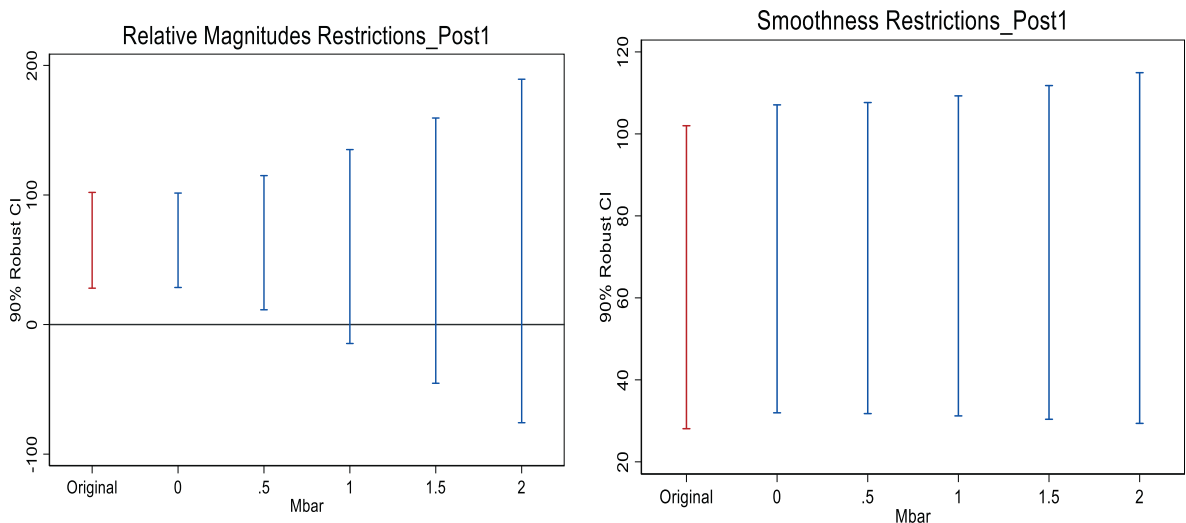


Fig. A5. Sensitivity test chart of parallel trend in the first year after policy implementation (The dependent variable is Margin).

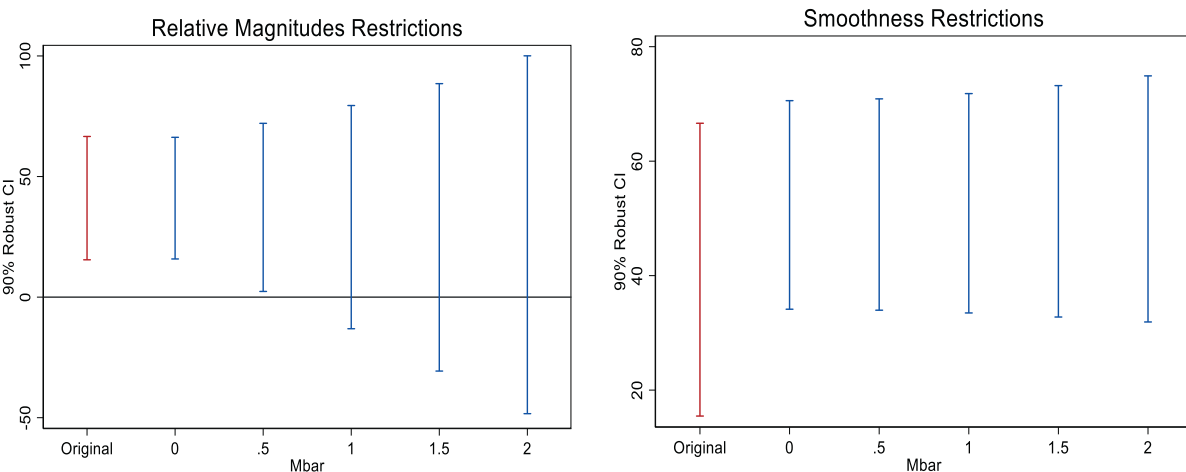


Fig. A6. Parallel trend hypothesis sensitivity test (The dependent variable is All_Pricing).

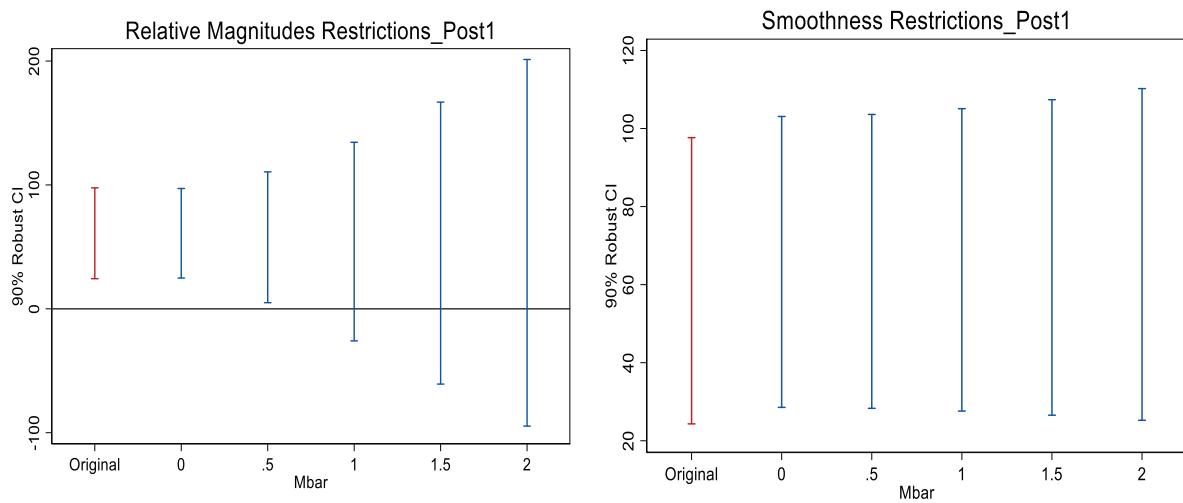


Fig. A7. Sensitivity test chart of parallel trend in the first year after policy implementation (The dependent variable is All_Pricing).

Table A5
Results of time placebo.

	(1)	(2)	(3)
	advanced by one year	advanced by two year	advanced by three years
	All_Pricing	All_Pricing	All_Pricing
<i>DID_1</i>	5.964 (6.694)		
<i>DID_2</i>		−0.116 (5.953)	
<i>DID_3</i>			3.802 (4.954)
<i>Pre_1</i>	35.133*** (7.028)		
<i>Pre_2</i>		36.422*** (6.969)	
<i>Pre_3</i>			35.794*** (6.965)
<i>Treat</i>	−56.946*** (12.999)	−56.415*** (12.906)	−57.856*** (12.661)
<i>_cons</i>	−97.733*** (33.285)	−97.685*** (33.296)	−97.298*** (33.227)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	31,018	31,018	31,018
<i>R²</i>	0.556	0.556	0.556

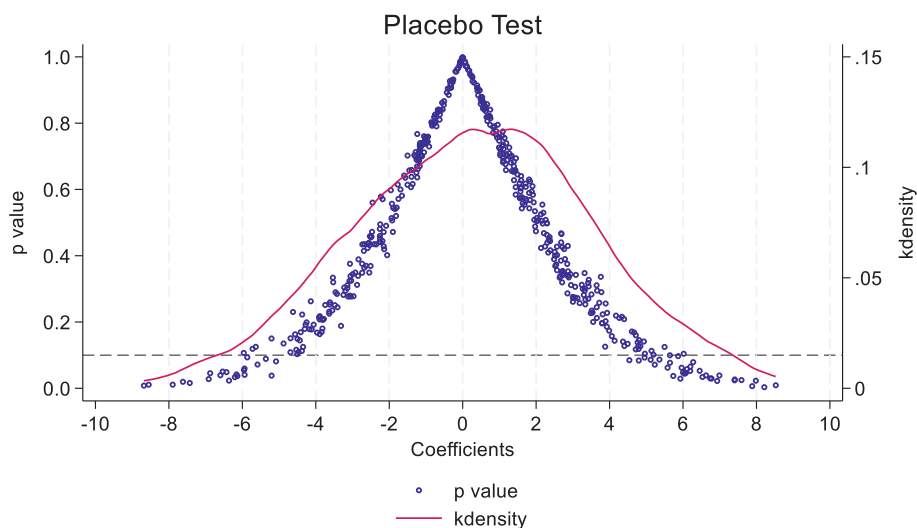


Fig. A8. Results of placebo test.

Table A6
Results of other robustness tests.

	(1)	(2)	(3)	(4)
	Bank_syndicate Fixed Effect	Borrower_country Fixed Effect	Ln(margin)	PSM
	All_Pricing	All_Pricing	Ln_All_Pricing	All_Pricing
<i>DID</i>	19.460*** (5.528)	16.434*** (5.622)	0.141*** (0.027)	26.962*** (6.767)
<i>Treat</i>	−80.498*** (12.742)	−58.574*** (9.416)	−0.341*** (0.035)	−57.804*** (13.382)
<i>Post</i>	32.198*** (5.021)	38.524*** (4.462)	0.130*** (0.030)	19.204** (8.465)
<i>_cons</i>	−118.284*** (29.808)	−157.828*** (29.542)	4.072*** (0.126)	−107.378*** (37.080)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>Bank_syndicate</i>	YES	YES	NO	NO
<i>Borrower_country</i>	NO	YES	NO	NO
<i>N</i>	31,018	31,018	31,018	27,482
<i>R</i> ²	0.647	0.667	0.638	0.572

Table A7
Results of different clustering of errors test.

	(1)	(2)	(3)	(4)
	Cluster by Bank & Year	Cluster by Borrower & Year	Cluster by SIC Code & Year	Jackknife
	All_Pricing	All_Pricing	All_Pricing	All_Pricing
<i>DID</i>	29.665*** (6.382)	29.665* (15.857)	29.665* (16.158)	29.665*** (6.751)
<i>Treat</i>	−57.887*** (13.770)	−57.887* (32.178)	−57.887* (33.405)	−57.887*** (13.839)
<i>Post</i>	30.697*** (6.933)	30.697* (16.539)	30.697* (16.757)	30.697*** (7.007)
<i>_cons</i>	−100.108*** (29.348)	−100.108 (80.455)	−100.108 (79.160)	−100.108*** (32.399)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>N</i>	31,018	31,018	31,018	31,018
<i>R</i> ²	0.557	0.557	0.557	0.557

Table A8

The impact of borrowers' debt-paying ability.

	(1)	(2)	(3)
	All Pricing	All Pricing	All Pricing
<i>DID</i> × <i>IntCoverage</i>	−23.697*** (8.513)		
<i>DID</i> × <i>Z_score</i>		−19.832*** (3.084)	
<i>DID</i> × <i>Rating</i>			3.530*** (1.322)
<i>IntCoverage</i>	−15.832*** (1.447)		
<i>Z_score</i>		−1.513** (0.710)	
<i>Rating</i>			2.743*** (0.740)
<i>DID</i>	69.456*** (20.095)	92.349*** (13.293)	10.913 (18.675)
<i>treat</i>	−44.118*** (13.705)	−57.958*** (12.550)	−79.013** (36.692)
<i>post</i>	31.911*** (7.330)	41.239*** (7.173)	44.256*** (14.645)
<i>_cons</i>	−75.552** (32.424)	−163.233*** (32.913)	−132.760*** (42.381)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	30,325	28,941	15,223
<i>R</i> ²	0.577	0.577	0.578

Table A9

The role of green banks in syndicated loans.

	(1)	(2)	(3)
	All Pricing	All Pricing	All Pricing
<i>DID</i> × <i>Green_bank</i>	15.242** (6.815)		
<i>DID</i> × <i>Green_syndicate</i>		7.715*** (1.439)	
<i>DID</i> × <i>Green_part</i>			1.485*** (0.441)
<i>DID</i>	26.641*** (7.128)	23.308*** (6.982)	22.881*** (7.280)
<i>Green_bank</i>	−1.398 (2.312)		
<i>Green_syndicate</i>		0.485 (0.635)	
<i>Green_part</i>			−28.430* (14.776)
<i>Treat</i>	−57.729*** (13.055)	−57.077*** (13.010)	−55.886*** (12.979)
<i>Post</i>	30.658*** (6.768)	33.322*** (7.151)	29.903*** (6.991)
<i>_cons</i>	−100.512*** (33.025)	−97.648*** (34.552)	−108.552*** (33.296)
<i>Controls</i>	YES	YES	YES
<i>Industry</i>	YES	YES	YES
<i>Year</i>	YES	YES	YES
<i>N</i>	31,018	31,018	31,018
<i>R</i> ²	0.557	0.558	0.558

Table A10

The difference between biodiversity Transition risk and physical risk in bank loan pricing.

	(1)	(2)	(3)	(4)
	Measure reference Coqueret et al. (2025)		Measure reference Giglio et al. (2023)	
	All Pricing	All Pricing	All Pricing	All Pricing
<i>DID_Transition</i>	36.086*** (4.253)		16.060*** (4.567)	
<i>Treat_Transition</i>	26.824** (11.554)		−15.860*** (5.451)	
<i>DID_Physical</i>		17.985*** (4.873)		7.001* (3.773)
<i>Treat_Physical</i>		−25.227*** (9.350)		−5.418 (6.521)
<i>Post</i>	27.261*** (7.127)	21.207*** (7.083)	34.043*** (6.944)	31.949*** (6.991)
<i>_cons</i>	−101.864*** (32.833)	−128.058*** (33.614)	−111.198*** (33.063)	−109.780*** (32.903)
<i>Controls</i>	YES	YES	YES	YES
<i>Industry</i>	YES	YES	YES	YES
<i>Year</i>	YES	YES	YES	YES
<i>N</i>	31,018	31,018	30,960	30,960
<i>R</i> ²	0.556	0.557	0.555	0.555

Table A11

The impact of different legal systems on the pricing of biodiversity risk in loans.

	(1)	(2)
	Common law	Civil law
	All Pricing	All Pricing
<i>DID</i>	28.298*** (6.563)	111.668 (79.043)
<i>Treat</i>	−36.253*** (10.479)	−128.002*** (23.522)
<i>Post</i>	40.198*** (6.611)	−54.041** (25.277)
<i>_cons</i>	−56.720* (33.192)	248.426** (99.748)
<i>Controls</i>	YES	YES
<i>Industry</i>	YES	YES
<i>Year</i>	YES	YES
<i>N</i>	28,790	2228
<i>R</i> ²	0.544	0.942

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Acharya, V. V., Berger, A. N., & Roman, R. A. (2018). Lending implications of U.S. bank stress tests: Costs or benefits? *Journal of Financial Intermediation*, 34, 58–90.

Adamolekun, G. (2024). Firm biodiversity risk, climate vulnerabilities, and bankruptcy risk. *Journal of International Financial Markets Institutions and Money*, 97, Article 102075.

Ahmad, M. F., & Karpuz, A. (2024). Beyond climate change risk: Biodiversity and corporate cash holdings. *Economics Letters*, 236, Article 111608.

Alessi, L., Di Girolamo, E. F., Pagano, A., & Giudici, M. P. (2024). Accounting for climate transition risk in banks' capital requirements. *Journal of Financial Stability*, 101269.

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4), 589–609.

Bach, T. N., Hoang, K., & Le, T. (2025). Biodiversity risk and firm performance: Evidence from US firms. *Business Strategy and the Environment*, 34(1), 1113–1132.

Badoer, D. C., & James, C. M. (2016). The determinants of long-term corporate debt issuances. *The Journal of Finance*, 71(1), 457–492.

Bassen, A., Buchholz, D., Lopatta, K., & Rudolf, A. R. (2024). Biodiversity management and stock price crash risk. *Business Strategy and the Environment*, 33(5), 4788–4805.

Beck, T., Levine, R., & Levkov, A. (2010). Big bad banks? The winners and losers from bank deregulation in the United States. *The Journal of Finance*, 65(5), 1637–1667.

Becker, A., Di Girolamo, F. E., & Rho, C. (2025). Loan pricing and biodiversity exposure: Nature-related spillovers to the financial sector. *Research in International Business and Finance*, 75, Article 102724.

Berger, A. N., Hasan, I., & Zhou, M. (2010). The effects of focus versus diversification on bank performance: Evidence from Chinese banks. *Journal of Banking & Finance*, 34(7), 1417–1435.

Bolton, P., & Kacperczyk, M. (2023). Global pricing of carbon-transition risk. *The Journal of Finance*, 78(6), 3677–3754.

Bose, S., Khan, H. Z., & Monem, R. M. (2021). Does green banking performance pay off? Evidence from a unique regulatory setting in Bangladesh. *Corporate Governance: An International Review*, 29(2), 162–187.

Cabernard, L., Pfister, S., & Hellweg, S. (2024). Biodiversity impacts of recent land-use change driven by increases in Agri-food imports. *Nature Sustainability*, 1–13.

Cahn, C., Girotti, M., & Salvade, F. (2024). Credit ratings and the hold-up problem in the loan market. *Management Science*, 70(3), 1810–1831.

Chava, S., & Roberts, M. R. (2008). How does financing impact investment? The role of debt covenants. *The Journal of Finance*, 63(5), 2085–2121.

Chen, I. J., Hasan, I., Lin, C. Y., & Nguyen, T. N. V. (2021). Do banks value borrowers' environmental record? Evidence from financial contracts. *Journal of Business Ethics*, 174, 687–713.

Chen, T. K., Liao, H. H., & Kuo, H. J. (2013). Internal liquidity risk, financial bullwhip effects, and corporate bond yield spreads: Supply chain perspectives. *Journal of Banking & Finance*, 37(7), 2434–2456.

- Chen, Y., Yin, L., Hu, S., & Li, X. (2025). Biodiversity conservation and corporate environmental information disclosure: Evidence from a quasi-natural experiment in China. *International Review of Financial Analysis*, 102, Article 104069.
- Cherief, A., Sekine, T., & Stagnol, L. (2025). A novel nature-based risk index: Application to acute risks and their financial materiality on corporate bonds. *Ecological Economics*, 228, Article 108435.
- Cooper, H. D., Mumba, M., Dhyani, S., Shen, X., Ma, K., Wynberg, R., ... Rodríguez, J. P. (2023). Priorities for progress towards sustainable development goal 15 'life on land'. *Nature Ecology & Evolution*, 7(10), 1570–1575.
- Coqueret, G., Giroux, T., & Zerbib, O. D. (2025). The biodiversity premium. *Ecological Economics*, 228, Article 108435.
- Delis, M. D., Greiff, K. D., Iosifidi, M., & Ongena, S. (2024). Being stranded with fossil fuel reserves? Climate policy risk and the pricing of bank loans. *Financial Markets, Institutions & Instruments*, 33(3), 239–265.
- Diamond, D. W. (1984). Financial intermediation and delegated monitoring. *The Review of Institutional Economics*, 51(3), 393–414.
- Diniz-Maganini, N., Rasheed, A. A., & Yasar, M. (2023). Legal systems and stock market efficiency: An empirical analysis of stock indices around the world. *Journal of Institutional Economics*, 19(4), 459–477.
- Dothan, M. (2010). Costs of financial distress and interest coverage ratios. *Journal of Financial Research*, 29(2), 147–162.
- Duffie, D., Saita, L., & Wang, K. (2007). Multi-period corporate default prediction with stochastic covariates. *Journal of Financial Economics*, 83(3), 635–665.
- Ehlers, T., Packer, F., & De Greiff, K. (2022). The pricing of carbon risk in syndicated loans: Which risks are priced and why? *Journal of Banking & Finance*, 136, Article 106180.
- El Ouadghiri, I., Kaabia, O., Peillex, J., Platania, F., & Hernandez, C. T. (2025). Attention to biodiversity and stock returns. *International Review of Financial Analysis*, 97, Article 103855.
- Ergungor, O. E. (2004). Market-vs. bank-based financial systems: Do rights and regulations really matter? *Journal of Banking & Finance*, 28(12), 2869–2887.
- Erten, I., & Ongena, S. (2024). Do banks price environmental risk? Only when local beliefs are binding!. Only When Local Beliefs are Binding. Swiss Finance Institute Research Paper No. 24–40, (Available at SSRN: 4932511).
- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fard, A., Javadi, S., & Kim, I. (2020). Environmental regulation and the cost of bank loans: International evidence. *Journal of Financial Stability*, 51.
- Ferdous, L. T., Atawnah, N., Yeboah, R., & Zhou, Y. (2024). Firm-level climate risk and accounting conservatism: International evidence. *International Review of Financial Analysis*, 95, Article 103511.
- Flammer, C., Giroux, T., & Heal, G. M. (2025). Biodiversity finance. *Journal of Financial Economics*, 164, Article 103987.
- Fu, C., Steckbauer, A., Mann, H., & Duarte, C. M. (2024). Achieving the Kunming-Montreal global biodiversity targets for blue carbon ecosystems. *Nature Reviews Earth and Environment*, 5(7), 538–552.
- Garel, A., Romec, A., Sautner, Z., & Wagner, A. F. (2024). Do investors care about biodiversity? *Review of Finance*, 28(4), 1151–1186.
- Ge, W., Kim, J. B., & Song, B. Y. (2012). Internal governance, legal institutions and bank loan contracting around the world. *Journal of Corporate Finance*, 18(3), 413–432.
- Giglio, S. A. K. T., Stroebel, J., & Zeng, X. (2023). Biodiversity risk. *NBER working paper* 31137.
- Graff, M. (2008). Law and finance: Common law and civil law countries compared—An empirical critique. *Economica*, 75(297), 60–83.
- Green, J. M., Croft, S. A., Durán, A. P., Balmford, A. P., Burgess, N. D., Fick, S., ... West, C. D. (2019). Linking global drivers of agricultural trade to on-the-ground impacts on biodiversity. *Proceedings of the National Academy of Sciences*, 116(46), 23202–23208.
- Guo, S., & Zhang, Z. (2023). Green credit policy and total factor productivity: Evidence from Chinese listed companies. *Energy Economics*, 128, Article 107115.
- Hadji-Lazaro, P., Salin, M., Svartzman, R., et al. (2024). Biodiversity loss and financial stability as a new frontier for central banks: An exploration for France. *Ecological Economics*, 223, Article 108246.
- He, F., Duan, L., Lucey, B., & Hao, J. (2025). Biodiversity risk or climate risk? Which reduces corporate ESG rating divergence. *International Review of Financial Analysis*, 104302.
- Hrazdil, K., Anginer, D., Li, J., & Zhang, R. (2023). Climate reputation and bank loan contracting. *Journal of Business Ethics*, 192(4), 875–896.
- Huang, H. H., Kerstein, J., Wang, C., & Wu, F. (2022). Firm climate risk, risk management, and bank loan financing. *Strategic Management Journal*, 43(13), 2849–2880.
- Huang, Y., Créti, A., Jiang, B., & Sanin, M. E. (2024). Biodiversity risk, firm performance, and market mispricing. Available at SSRN 4765039.
- Javadi, S., & Masum, A.-A. (2021). The impact of climate change on the cost of bank loans. *Journal of Corporate Finance*, 69, Article 102019.
- Jawadi, F., Rozin, P., & Cheffou, A. I. (2024). Toward green central banking: Proposing an augmented Taylor rule. *Energy Economics*, 134.
- Johnson, T. F., Beckerman, A. P., Childs, D. Z., Webb, T. J., Evans, K. L., Griffiths, C. A., ... Freckleton, R. P. (2024). Revealing uncertainty in the status of biodiversity change. *Nature*, 628(8009), 788–794.
- Kalhor, M. R., & Kyaw, K. (2024). Manage biodiversity risk exposure? *Finance Research Letters*, 61, Article 104989.
- Kandrac, J., & Schlusche, B. (2021). The effect of bank supervision and examination on risk taking: Evidence from a natural experiment. *The Review of Financial Studies*, 34(6), 3181–3212.
- Lampert, A. (2019). Over-exploitation of natural resources is followed by inevitable declines in economic growth and discount rate. *Nature Communications*, 10(1), 1419.
- Leland, H. E., & Pyle, D. H. (1977). Informational asymmetries, financial structure, and financial intermediation. *The Journal of Finance*, 32(2), 371–387.
- Li, Y., Liu, X., Canil, J., & Cheong, C. S. (2025). Biodiversity risk and firm efficiency. *Finance Research Letters*, 71, Article 106414.
- Liang, C., Yang, J., Shen, L., & Dong, D. (2024). The role of biodiversity risk in stock price crashes. *Finance Research Letters*, 67.
- Liu, X., Li, Y., & Krishnamurti, C. (2025). Biodiversity risk and seasoned equity offerings. *Economics Letters*, 247, Article 112214.
- Liu, Z., He, S., Men, W., & Sun, H. (2024). Impact of climate risk on financial stability: Cross-country evidence. *International Review of Financial Analysis*, 92, Article 103096.
- Ma, F., Wu, H., & Zeng, Q. (2024). Biodiversity and stock returns. *International Review of Financial Analysis*, 95, 103386.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of Finance*, 29(2), 449–470.
- Mundaca, L., & Heintze, J. N. (2024). Banking on ecosystem services. *Ecological Economics*, 224, Article 108284.
- Nicholson, E., Andrade, A., Brooks, T. M., Driver, A., Ferrer-Paris, J. R., Grantham, H., ... Obura, D. (2024). Roles of the red list of ecosystems in the Kunming-Montreal global biodiversity framework. *Nature Ecology & Evolution*, 8(4), 614–621.
- Pi, T., Jiao, L., Zhou, Y., & Shi, J. (2025). Can biodiversity risk improve firm ESG performance? Empirical evidence from China. *Finance Research Letters*, 76, Article 106931.
- Reghezza, A., Altunbas, Y., Marques-Ibanez, D., Rodriguez d'Acari, C., & Spaggiari, M. (2022). Do banks fuel climate change? *Journal of Financial Stability*, 62, Article 101049.
- Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2), 2218–2244.
- Shears, E., Meckling, J., & Finnegan, J. J. (2025). How central banks manage climate and energy transition risks. *Nature Energy*, 1–9.
- Soylomezgil, S., & Uzmanoglu, C. (2024). Biodiversity risk in the corporate bond market. Available at SSRN 4721219.
- Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3), 393–410.
- Stroebel, J., & Wurgler, J. (2021). What do you think about climate finance? *Journal of Financial Economics*, 142(2), 487–498.
- Swinfield, T., Shrikanth, S., Bull, J. W., Madhavapeddy, A., & Zu Ermgassen, S. O. (2024). Nature-based credit markets at a crossroads. *Nature Sustainability*, 7(10), 1217–1220.
- Tian, Y., & Chen, H. (2025). Eco-innovation under pressure: How biodiversity risks shape corporate sustainability strategies. *Finance Research Letters*, 73, Article 106615.
- Trinh, H. H. (2023). Biodiversity risks and corporate investment. Available at SSRN 4587333.
- Wang, J., Qiang, H., Liang, Y., Huang, X., & Zhong, W. (2024). How carbon risk affects corporate debt defaults: Evidence from Paris agreement. *Energy Economics*, 129, Article 107275.
- Wang, S., Zhang, R., Wan, L., & Chen, J. (2023). Has central government environmental protection interview improved air quality in China? *Ecological Economics*, 206, Article 107750.
- Wang, Y. (2025). The silent cost of biodiversity loss: Unveiling its impact on institutional ownership. *International Review of Financial Analysis*, 103, Article 104075.
- WWF. (2024). *Living planet report 2024-A system in peril*. Gland, Switzerland: WWF.
- Xiong, W. W. (2023). *The world market price of biodiversity risk*. Available at SSRN 5043322.
- Yang, G., & Li, Y. (2025). Research on biodiversity risk and corporate short-term debt for long-term use under disclosure regulation. *Finance Research Letters*, 77, 107049.
- Yang, Y., Tilman, D., Jin, Z., Smith, P., Barrett, C. B., Zhu, Y. G., ... Zhuang, M. (2024). Climate change exacerbates the environmental impacts of agriculture. *Science*, 385(6713), 3747.
- Zhang, D., Bai, D., & Wang, Y. (2024). Green vs. brown: Climate risk showdown - who's thriving, who's diving? *Journal of International Money and Finance*, 149.